

A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

ABSTRACT. A countable discrete group is MF if it embeds in the unitary group of a matrix quotient $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$; equivalently, if it admits finite-dimensional approximate unitary representations, asymptotically multiplicative in operator norm, that separate every nontrivial element from the identity. We construct an explicit finitely presented group E that is not MF, refuting the conjecture that every countable group is MF.

More generally, let Γ be a finitely presented property-(T) group with a proper injective endomorphism α . From (Γ, α) we build a finitely presented group with a nontrivial central involution w such that $\Theta(w) = 1$ for every homomorphism Θ to $\mathcal{U}(\mathcal{Q})$. In every operator-norm asymptotic representation, one-sided conjugation preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm; if $\Theta(w) \neq 1$, compression to the (-1) -spectral projection of $\Theta(w)$ contradicts this. Taking $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and $\alpha(v, A) = (2v, A)$ gives E , on eight generators and forty-one relators.

The group E is sofic, hence hyperlinear, so neither soficity nor hyperlinearity implies the MF property. Its reduced C^* -algebra $C_r^*(E)$ is separable, exact and stably finite but not MF. The quotient $E/\langle w \rangle$ is not MF either, so $\{1, w\}$ is a proper subgroup of the kernel common to all homomorphisms $E \rightarrow \mathcal{U}(\mathcal{Q})$.

2020 *Mathematics Subject Classification*. Primary 46L05; Secondary 20F05, 22D25, 20E26.

Key words and phrases. MF groups, approximate unitary representations, property (T), group C^* -algebras, sofic groups, Clifford groups.

CONTENTS

1. Introduction	3
2. Matrix quotients	6
3. One-sided conjugation in matrix models	8
4. The central involution obstruction	11
5. The Kazhdan–Clifford construction	12
6. A finitely presented non-MF group	13
7. Consequences of the example	16
Supplementary material	18
S1. The base, and a smaller presentation	18
S2. Nontriviality of w	18
S3. The MF equivalences	20
S4. Weights and intertwiners	22
S5. Obstructions from a one-sided conjugation datum	23
S6. The finite-dimensional obstruction	28
S7. The cyclic comparison	29
S8. The scaling family	31
S9. The MF residual	31
S10. Orbit collapse	33
S11. Group algebras, soficity, and permanence	37
S12. Marked limits, quasi-identities, and undecidability	41
S13. Limitations of the operator-norm method	43
S14. The maximal group C^* -algebra	46
Acknowledgements	47
References	47

1. INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP].

The result.

Theorem A (an explicit finitely presented non-MF group). *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

The defining relations make w central, and with $c^2 = 1$ centrality forces $w^2 = 1$.

- (1) *$w \neq 1$ in E , so w is a central involution;*
- (2) *for every sequence (d_n) of positive integers and every homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C_{\max}^(E)$ nor $C_r^*(E)$ is an MF C^* -algebra.*

Outline of the proof. Operator norm is the approximation norm throughout, and the normalized Hilbert–Schmidt norm appears only as the norm of the Hilbert space on which the conjugation representation acts.

Regard $M_d(\mathbb{C})$ as the Hilbert space $L^2(M_d(\mathbb{C}), \text{tr}_d)$, on which conjugation by a unitary acts unitarily. For unitaries u_1, u_2 , uniformly in d ,

$$\|\text{Ad } u_1 - \text{Ad } u_2\|_{B(L^2(M_d(\mathbb{C}), \text{tr}_d))} \leq 2 \|u_1 - u_2\|,$$

so an operator-norm asymptotic representation induces one by conjugation on the Hilbert–Schmidt spaces.

Let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and let $t, c \in H$ satisfy

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

In a norm ultraproduct B_ω of the blocks, the classes $\pi(g) = [\text{Ad } U_n(g)]_\omega$ define a unitary representation of H . By property (T), the spectrum of the Hermitian average h of the generators is bounded away from 1 off the invariant vectors, so the spectral projection of h at 1 lies in B_ω and projects onto the $\iota(\Gamma)$ -invariant vectors; call it P . Writing $V = \pi(t)$, the inclusion above implies $P \leq VPV^*$; the two projections are Murray–von Neumann equivalent, and B_ω is finite, so $P = VPV^*$. Hence conjugation by t preserves

the asymptotic commutant of $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for $a \in \Gamma$. Since c commutes with $\iota(\Gamma)$, its lifts lie in that asymptotic commutant, so by Theorem 3.4 $\| [U_n(d), U_n(\iota(\gamma))] \|_2 \rightarrow 0$ for every $\gamma \in \Gamma$. Consequently, in every operator-norm asymptotic representation (U_n) of H ,

$$\|U_n(u) - 1\|_2 \rightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with $\Theta(H)$, and rounding the coordinate averages of unitary lifts represents q by projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero for infinitely many n . Compressing to q_n , and giving each corner its own normalized trace, yields an operator-norm asymptotic representation of H in which w goes to -1 . There the display above gives $\|U_n(u)^2 - 1\|_2 \rightarrow 0$, while $\|-1 - 1\|_2 = 2$. Hence $\Theta(w) = 1$, and if $w \neq 1$ then H is not MF (Theorem 4.1).

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α , let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the direct limit of α the cosets $t\Gamma$ and $at\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. Let the ambient group permute the generators of a Clifford group indexed by that coset space, and send c to the generator at the base coset; then d and ada^{-1} map to the generators at those two cosets. Distinct Clifford generators anticommute, so u maps to a product of two anticommuting involutions and $w = u^2$ to -1 (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I)$$

gives the group E of Theorem A.

The same one-sided inclusion has exact analogues, which is where the mechanism is easiest to see. For a homomorphism π to a finite group, $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$, since the left side is a conjugate of the right side contained in it and the two are finite of the same cardinality; so $\pi(tct^{-1})$ centralizes $\pi(\Gamma)$ and every commutator $[tct^{-1}, a]$ with $a \in \Gamma$ lies in $\ker \pi$. For a finite-dimensional representation over any field the same conclusion holds with the commutant C of $\pi(\Gamma)$ in place of the subgroup: the inclusion gives $C \subseteq \pi(t)C\pi(t)^{-1}$ between subspaces of equal finite dimension, hence equality, and $\pi(c) \in C$ then puts $\pi(tct^{-1})$ in C (Theorem 3.3). In $\mathcal{Q}$ neither cardinality nor dimension is available, and finiteness of the norm ultraproduct plays their part.

Only the subgroup $\iota(\Gamma)$ is assumed to have property (T). The group E surjects onto $\mathbb{Z}$, by sending t to 1 and every other generator to 0, so E itself does not have property (T).

The reduced group C^* -algebra $C_r^*(E)$ is separable and stably finite but not MF (Theorem C); it is also exact. The group E is sofic (Theorem D), so

neither soficity nor hyperlinearity implies the MF property. The quotient $E/\langle w \rangle$ is not MF either, so $\{1, w\}$ is a proper subgroup of the MF residual of E (Theorem B). The non-MF argument by itself decides neither soficity nor nonsocicity of E (Section S13).

Relation to prior work. The corresponding C^* -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^* -algebra was known not to be MF [BK, CDE]. Their natural candidate was $C_r^*(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group C^* -algebra fails to be MF [Sh].

In his 2018 ICM address, Thom asked for which Schatten p -norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky–Oppenheim [LuO], and the case $p = 1$ by Bachner–Dogon–Lubotzky [BDL]. Three approximation problems were left by that work: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), and MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; at the time of writing the hyperlinear problem is open. Our construction begins, as those arguments do, with a relation $t\Gamma t^{-1} \subseteq \Gamma$ for a property-(T) subgroup Γ . Here the approximation is by unitaries in operator norm, and conjugation by t preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. Throughout, “MF” is the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE], in the form used by Shulman and by Bachner–Dogon–Lubotzky [Sh, BDL]; Section 2 fixes it and its equivalent formulations.

The closest analogue is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and finiteness of the norm ultraproduct as a C^* -algebra forces the projection onto the $\iota(\Gamma)$ -fixed subspace to equal its unitary conjugate (Theorem 3.4).

Slofstra’s hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. He also considers the elements sent to the identity

by every exact finite-dimensional representation, and those sent there by every approximate one [Slo17, Definitions 2.5–2.7]; the two subgroups they form are the exact and the approximate analogues of the MF residual used below. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz’s marked $BS(2, 3)$ example is the context for the cyclic comparison of Section S7 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section S5 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5], and handle a finite normal subgroup by a character-isotypic corner [BDL, proof of Proposition 1.6]; Theorem S5.3 combines that corner with one-sided conjugation of a property-(T) subgroup.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

The group property is weaker than the algebra property. For the algebra one asks for an embedding $C_r^*(E) \hookrightarrow \mathcal{Q}$ of C^* -algebras; for the group, only for an injective homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$, which need not extend to the algebra. Failure of MF for a group C^* -algebra therefore leaves open whether the group itself embeds. For E , Theorem A rules out both.

2. MATRIX QUOTIENTS

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements $[g, h] = ghg^{-1}h^{-1}$; for operators $[x, y]_- = xy - yx$. A C^* -homomorphism is unital only when this is stated; in particular an MF embedding may be nonunital. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q \in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace, and where the trace has to be named we write $\|x\|_{2, \text{tr}_r}$ for the same quantity. The unnormalized trace is $\text{Tr} = r \cdot \text{tr}_r$ on $M_r(\mathbb{C})$. We use without comment: $\|x\|_2 \leq \|x\|$; $|\text{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in \mathbb{N}}$ of positive integers, write

$$(1) \quad \mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}),$$

where the product is the bounded C^* -product and $\bigoplus_n$ is the c_0 -sum, the ideal of sequences with $\|x_n\| \rightarrow 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q: \prod_n M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ for the quotient map. Following

Blackadar–Kirchberg [BK], a separable C^* -algebra is MF if it embeds into an algebra of the form (1).

The following lifting lemmas are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 2.1 (lifting). *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n) ; unitarity of $q((x_n))$ gives $\|x_n^*x_n - 1\| \rightarrow 0$. Once $\|x_n^*x_n - 1\| \leq \frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \leq 2\|x_n\|\|x_n^*x_n - 1\| \rightarrow 0$ by continuous functional calculus; set $u_n = 1$ at the finitely many remaining indices. $\square$

Write

$$(2) \quad \mathcal{N} = \{(u_n) \in \prod_n \mathcal{U}(d_n) : \|u_n - 1\| \rightarrow 0\},$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient). *For every sequence (d_n) , with $\mathcal{N}$ as in (2), the map*

$$\kappa_{(d_n)}: \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. A homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ is therefore equivalent to the following data: a sequence of unitaries for each element of H , multiplicative up to operator-norm errors tending to 0, taken modulo sequences converging to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective. Surjectivity is Lemma 2.1. $\square$

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n: H \rightarrow \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \rightarrow 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is MF if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

Proposition 2.3 (Equivalent formulations of the MF property). *For every countable group, the group-MF definition above — an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

Section S3 defines the approximate unitary representations and proves the proposition. Two stronger conventions occur in the literature and are not used here: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element, and

the purely matricial field (PMF) property of [MdLS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. ONE-SIDED CONJUGATION IN MATRIX MODELS

In Theorem 3.3 the commutant C of the base inside a finite-dimensional endomorphism algebra satisfies $C \subseteq \pi(t)C\pi(t)^{-1}$, and equal finite dimension forces equality. In the proof of Theorem 3.4 the projection P onto the Kazhdan fixed space satisfies $P \leq VPV^*$ inside a norm ultraproduct, by $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$; the two projections are Murray–von Neumann equivalent and the ultraproduct is finite, so $P = VPV^*$.

Lemma 3.1 (comparison in a finite algebra). *Let A be a finite unital C^* -algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \quad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1 - q)$, and $\sigma^*\sigma = q + (1 - q) = 1$, so σ is an isometry. Finiteness makes σ unitary, and $\sigma\sigma^* = p + (1 - q) = 1$ gives $p = q$. $\square$

Lemma 3.2 (finiteness of a norm quotient of matrix blocks). *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_c = \prod_n B(K_n) / \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0 -sum. Then B_c is a finite C^ -algebra: every $\sigma \in B_c$ with $\sigma^*\sigma = 1$ satisfies $\sigma\sigma^* = 1$.*

Proof. Lift σ to a bounded family (σ_n) . Then $\|\sigma_n^*\sigma_n - 1\| \rightarrow 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^*\sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary $w_n = \sigma_n(\sigma_n^*\sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \rightarrow 0$. Hence σ is the class of (w_n) , and $w_n w_n^* = 1$ at every such n gives $\sigma\sigma^* = 1$. $\square$

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n , and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$. The proof above applies to B_ω with “for all large n ” replaced by “on a set belonging to ω ”, so B_ω is finite too. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n\xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\text{ran } P \subseteq \text{ran } Q$ makes QP and P act identically on K_ω , so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi: H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such π is trivial on the subgroup $\mathfrak{D}(H, \Gamma)$ of (9).

Proof. Let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\text{End}_k(V)$, and write $\Phi(x) = \pi(t)x\pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\text{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1}x\pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = \pi(t)^{-1}\pi(t\gamma t^{-1})x\pi(t) = \pi(t)^{-1}x\pi(t\gamma t^{-1})\pi(t) = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(3) \quad C = \Phi(C) = \pi(t)C\pi(t)^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ , so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (3). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$ and

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1.$$

Section S6 proves the subgroup-valued assertion from the same calculation. $\square$

In E , every finite-dimensional linear representation over every field, and hence every homomorphism to a finite group, maps the distinguished word to the identity (Corollary S6.1).

An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not invariant subspaces. Operator-norm control keeps the adjoint actions almost multiplicative on the Hilbert–Schmidt spaces, so the Kazhdan projection lies in the norm ultraproduct, where finiteness replaces the equality of dimensions used above. Property (T) cannot be dropped: the group E_{BS} of Section S7 has cyclic base, its distinguished word is trivial in every finite-dimensional representation over every field, and some homomorphism $E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$ maps that word to a nontrivial element.

Theorem 3.4 (one-sided conjugation preserves asymptotic commutants). *Let Γ and H be groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n: H \rightarrow \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm,

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma),$$

then the conjugated sequence does as well:

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma).$$

Proof. Suppose the conclusion fails: there are $\gamma_0 \in \Gamma$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ and let each group element act on it by conjugation, $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$. Operator-norm almost multiplicativity of U_n makes $g \mapsto \text{Ad } U_n(g)$ an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\cdot]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, B_ω is finite — every isometry in it is unitary — and an inclusion of ranges in K_ω is an inequality of projections in B_ω . Asymptotic multiplicativity makes

$$\pi: H \longrightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Property (T) provides a finite symmetric generating set $S \subseteq \Gamma$ with $1 \in S$ and a Kazhdan constant $\kappa \in (0, 1]$ for the pair (S, κ) . Let h be the Hermitian part of $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$ and let $\text{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\iota(\gamma))$. Both Fix and its orthocomplement are invariant under each unitary $\pi(\iota(s'))$, hence under h . On Fix , $h = 1$. The restriction of $\pi \circ \iota$ to $\text{Fix}^\perp$ has no nonzero invariant vector, so a unit vector $\xi \in \text{Fix}^\perp$ satisfies $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$ for some generator. For that generator $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$, while every other term is at most 1. Averaging gives $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$. Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection P of h at the isolated point 1 belongs to B_ω by continuous functional calculus and projects onto Fix .

The two projections agree. Let $V = \pi(s)$ and $Q = VPV^*$. Since $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$, every vector of Fix is fixed by each $\pi(s\iota(\gamma)s^{-1})$; and η is fixed by all $\pi(s\iota(\gamma)s^{-1})$ exactly when $V^*\eta \in \text{Fix}$. Hence $\text{ran } P = \text{Fix} \subseteq V\text{Fix} = \text{ran } Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [\xi_n]_\omega \in K_\omega$, where $\xi_n \in K_n$ is the vector of x_n ; the uniform operator-norm bound makes ξ well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\iota(\gamma))$ fixes ξ . Thus $\xi \in \text{Fix}$. Since $Q = P$, we obtain

$V\xi \in V\text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}$. The vector of $U_n(s)x_nU_n(s)^*$ is $\text{Ad } U_n(s)\xi_n$, so, again by unitary invariance,

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]\|_2 = \|(\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s)\xi_n\| \longrightarrow 0$$

along ω , for every $\gamma \in \Gamma$. For $\gamma = \gamma_0$ this contradicts $I \in \omega$. $\square$

4. THE CENTRAL INVOLUTION OBSTRUCTION

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, the projection $\frac{1}{2}(1 - \Theta(w))$ is nonzero, and compressing to it gives a matrix model of H in which w equals -1 and each block has its own normalized trace.

On the corner $q_n M_{d_n}(\mathbb{C}) q_n$ the operator norm is the restriction of the ambient one, so the compressed and polar-corrected maps remain operator-norm asymptotically multiplicative, while the trace is the corner's own normalized trace tr_{r_n} , not the restriction of tr_{d_n} . Hence $\|1 - (-1)\|_{2, \text{tr}_{r_n}} = 2$ for every value of r_n/d_n . Compression converts a normalized Hilbert–Schmidt bound into an operator-norm obstruction: padding an asymptotic representation with an identity block leaves every operator-norm quantity unchanged while sending every normalized Hilbert–Schmidt distance to zero (Section S13).

Theorem 4.1 (central involution obstruction). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for some $a \in \Gamma$. If $w := u^2$ is a nontrivial central involution of H , then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1) -spectral corner. Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, the Hermitian part of p_n is asymptotically idempotent, so its spectrum clusters at $\{0, 1\}$ and rounding at $\frac{1}{2}$ by continuous functional calculus produces genuine projections at operator-norm distance $o(1)$ from p_n ; centrality of w makes them asymptotically central for all of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Compressing every lift to $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ and polar-correcting the compressions — each is almost unitary in the corner, by asymptotic centrality — gives an operator-norm asymptotic representation $(W_{g,n})$ of H on nonzero blocks with

$$\|W_{w,n} + 1\| \longrightarrow 0.$$

Equip each corner block with its own normalized trace tr_{r_n} ; all estimates below are relative to r_n and independent of the ratio r_n/d_n .

The commutant on the corner. Since c centralizes $\iota(\Gamma)$, the sequence $(W_{c,n})$ centralizes $(W_{\iota(\gamma),n})$ asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.4 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{d,n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \rightarrow 0$, and squaring gives $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$.

The contradiction. The relation $w = u^2$ gives $\|W_{u,n}^2 - W_{w,n}\| \rightarrow 0$, while $W_{w,n} \rightarrow -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\|W_{u,n}^2 + 1\|_2 \rightarrow 0$ and $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$, so $\|2 \cdot 1\|_2 \rightarrow 0$, which is false. Thus $\Theta(w) = 1$. Since $w \neq 1$, every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ has nontrivial kernel, so H is not MF. $\square$

The same conclusion holds for an arbitrary finite normal subgroup of N_{conj} (Theorem S5.3), and for an arbitrary normal property-(T) subgroup $K \leq N_{\text{conj}}$, with a Kazhdan spectral projection in place of the two-element average; there K may be noncentral and torsion-free (Theorem S5.6).

5. THE KAZHDAN–CLIFFORD CONSTRUCTION

Theorem 5.1 (Kazhdan–Clifford construction). *Let Γ be a finitely presented group with property (T), let $\alpha: \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

and impose centrality of w . More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

Thus w commutes with a generating set of the entire resulting group. The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \rightarrow E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism $E(\Gamma, \alpha, a) \rightarrow \mathcal{U}(\mathcal{Q})$ maps w to 1, so $E(\Gamma, \alpha, a)$ is not MF.

Proof. Finite presentability follows from a relative finite presentation: start with the free product of Γ and the free group on t, c , and quotient by the finitely many displayed relations for a finite generating set of Γ .

The relation $w^2 = 1$ is redundant. Put $b = ada^{-1}$. Both d and b are involutions, $w = [d, b] = (db)^2$, and direct cancellation gives

$$(4) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

concretely. Let Γ_∞ be the direct limit of the system $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$, whose n th copy of Γ we call its n th level; injectivity of α passes to the limit, so the level-zero map embeds Γ in Γ_∞ . Applying α levelwise induces an injective endomorphism σ of Γ_∞ . It is surjective, hence an automorphism, because an element at level k equals σ of the same element at level $k+1$. Put $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$, with stable letter t implementing σ ; then $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$. Write $X = G/\Gamma$ for the coset space of the level-zero copy. Let G permute the Clifford generators $(e_x)_{x \in X}$ and map c to e_Γ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Thus $w = u^2$ is a nontrivial central involution, and by Theorem 4.1 every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, every such homomorphism has nontrivial kernel. $\square$

6. A FINITELY PRESENTED NON-MF GROUP

Specialize Theorem 5.1 to

$$(5) \quad \Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I).$$

The affine group Γ is finitely presented and has property (T) [BHV, Example 1.7.4(i)], and Section S1 identifies it with the presentation used below. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, it is proper because $e_1 \notin 2\mathbb{Z}^3$, and $a \notin \alpha(\Gamma)$ for the same reason. Theorem 5.1 therefore applies, and relative to Γ the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \mid \begin{array}{ll} t\gamma t^{-1} = \alpha(\gamma), & [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, & [w, g] = 1 \end{array} \right\rangle,$$

where S_Γ is a finite generating set of Γ , g runs over the generators, and $w = [tct^{-1}, a(tct^{-1})a^{-1}]$. Expanding a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear part turns this into the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$(6) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *base*

(7)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \left| \begin{array}{l} \text{the eight relations in (6),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right. \right\rangle.$$

Write $\iota: \mathcal{B} \rightarrow E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

Definition 6.1. Use the six base generators and the relations displayed in (7). Define

(8)

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{(B) all relations in (7),} \\ \text{(T) } tv_it^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C) } c^2 = 1, \\ [c, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W) } [w, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Family (B) is a presentation of the base; family (T) conjugates each base generator by t into the base; family (C) makes c an involution commuting with the base generators; and family (W) makes w central. Centrality together with $c^2 = 1$ implies $w^2 = 1$ by (4). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

Figure 1 is the configuration these two relations create: $d = tct^{-1}$ commutes with the conjugate copy $t\iota(\mathcal{B})t^{-1}$, while nothing constrains its commutator with $\iota(v_1)$, which lies outside that copy.

A Tietze reduction removes two generators and nine relators (Section S1).

Lemma 6.2 (the distinguished word is a square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

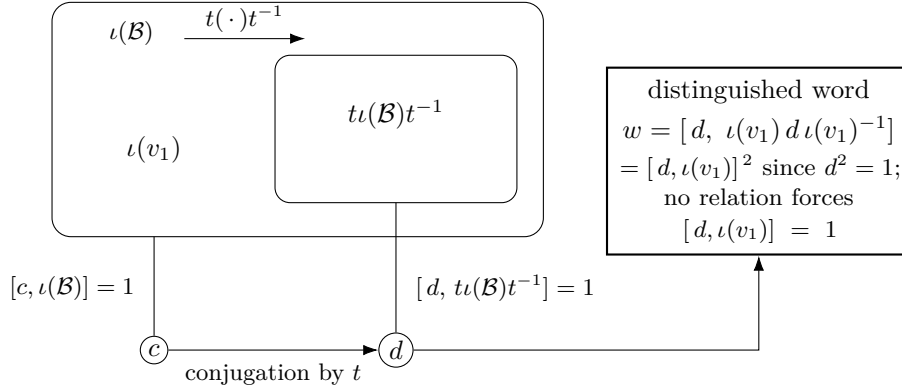


Figure 1: The one-sided conjugation configuration. The relations of (8) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[d, t\iota(\mathcal{B})t^{-1}] = 1$ for $d = tct^{-1}$; they do not give $[d, \iota(v_1)] = 1$, and $w = [d, \iota(v_1)d\iota(v_1)^{-1}] = [d, \iota(v_1)]^2$. Section S2 proves that the inclusion is strict and that $\iota(v_1)$ lies outside the conjugate copy, as drawn; Theorem 4.1 uses only the inclusion.

Proof of Theorem A. That $w \neq 1$ is Proposition S2.3.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.1) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι , the elements $t, c \in E$, and $a = v_1$. Its hypotheses are the displayed relators: Proposition S1.1 gives property (T) of $\mathcal{B}$; family (T) gives $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and family (C) gives $[c, \iota(\mathcal{B})] = 1$. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $a_1 = \iota(v_1)$ gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2, \quad u = [d, \iota(v_1)].$$

Centrality of w is a defining relator; with $c^2 = 1$ it gives $w^2 = 1$ by (4); and $w \neq 1$ as above. Hence $\Theta(w) = 1$.

An MF embedding of E would be an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, and every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1; so E is not MF. For the two algebras, the canonical maps $E \rightarrow \mathcal{U}(C_r^*(E))$ and $E \rightarrow \mathcal{U}(C_{\max}^*(E))$ are injective, the first because $g \mapsto \lambda_g$ is and the second because the left regular representation factors through $C_{\max}^*(E)$; and an injective, possibly nonunital $*$ -homomorphism $e: A \rightarrow \mathcal{Q}$ of a unital algebra becomes an injective homomorphism $\mathcal{U}(A) \rightarrow \mathcal{U}(\mathcal{Q})$ after *complement correction*, $u \mapsto e(u) + 1 - e(1)$. An MF embedding of either algebra would compose with these to give an injective homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. $\square$

Corollary 6.3 (uniform finite operator-norm obstruction). *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, for every $d \geq 1$, every map $\phi: E \rightarrow \mathcal{U}(d)$ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \dots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E , so by Lemma 2.2 its class is a homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, against Theorem A. $\square$

The proof is nonquantitative: it gives no bound for δ or for $|F_0|$. An effective modulus would need an effective Kazhdan constant for the base together with effective versions of the compression estimates.

7. CONSEQUENCES OF THE EXAMPLE

Throughout, $d = tct^{-1}$ and $u = [d, \iota(v_1)]$, so that $w = u^2$. Write $\text{Res}_{\text{MF}}(G)$ for the *MF residual* of a countable group G : the intersection of the kernels of all its homomorphisms to $\mathcal{U}(\mathcal{Q})$ (Definition S9.1). Thus G is MF exactly when $\text{Res}_{\text{MF}}(G) = 1$ (Proposition S9.3), and Theorem A says $w \in \text{Res}_{\text{MF}}(E)$ with $w \neq 1$.

The quotient by the central involution.

Theorem B (the quotient of E by $\langle w \rangle$). *The quotient $E/\langle w \rangle$ is not MF. The image of $[\iota(v_1), d]$, the inverse of the distinguished commutator u , is nontrivial there and lies in $\text{Res}_{\text{MF}}(E/\langle w \rangle)$; consequently $\{1, w\}$ is a proper subgroup of $\text{Res}_{\text{MF}}(E)$.*

If an involution is centralized by the conjugate copy of a Kazhdan subgroup, and its conjugates under that subgroup commute with one another, then every homomorphism to $\mathcal{U}(\mathcal{Q})$ identifies the entire conjugation orbit (Theorem S10.3). In E the pairwise commutators of those conjugates of d lie in $\langle w \rangle$, so in $E/\langle w \rangle$ the element d satisfies these hypotheses, and every homomorphism $E/\langle w \rangle \rightarrow \mathcal{U}(\mathcal{Q})$ maps u to the identity. Thus $E/\langle w \rangle$ is a second finitely presented non-MF group, and Theorem A does not determine $\text{Res}_{\text{MF}}(E)$ (Section S13).

A stably finite non-MF group algebra. Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on $C_r^*(E)$ is faithful, so $C_r^*(E)$ is stably finite (Lemma S11.1), independently of the MF question. Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group E is an explicit one (Theorem C). For $C_r^*(\text{SL}_3(\mathbb{Z}))$ and $C_r^*(\text{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\text{SL}_4(\mathbb{Z})$ [Mdis].

Theorem C (the reduced group algebra of E). *$C_r^*(E)$ is a unital separable C^* -algebra which has a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

Proof. $C_r^*(E)$ is unital and separable because E is countable, and by Lemma S11.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A. $\square$

The algebra is also exact, by the block normal form of E and permanence for free products and extensions (Section S11). The algebra is not nuclear: the canonical map $\mathcal{B} \rightarrow E$ is injective (Section S2), so E contains an infinite Kazhdan group and is nonamenable, whence $C_r^*(E)$ is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear C^* -algebra is quasidiagonal — therefore remains open.

A sofic example.

Theorem D (a finitely presented sofic non-MF group). *The group E of Definition 6.1 is sofic, hence hyperlinear. Consequently E is itself a finitely presented sofic group that is not MF, and a finitely presented hyperlinear group that is not MF. It is also sofic and not LEF, and in particular not residually finite.*

The proof is in Section S11.3. Let E_0 be the kernel of the exponent sum in t . A finite subset of E_0 involves finitely many conjugates of c and lies over a single level of the ascending chain, and the subgroup they generate embeds in a residually finite group; so E_0 is a directed union of such subgroups, hence LEF, and E is a split extension of E_0 by $\mathbb{Z}$, hence sofic by the Elek–Szabó amenable-extension theorem [ES06]. That last step is where LEF is lost: LEF groups are MF [CDE], so E_0 is LEF and E is not.

Filling in the anticommutation relations at every pair of distinct cosets maps E onto the Clifford group $W = \text{Cl}(X) \rtimes G$ built in the proof of Theorem 5.1, and carries w to the Clifford sign, so functoriality (Lemma S9.2) puts that sign in $\text{Res}_{\text{MF}}(W)$ and W is not MF either. Its defining extension has locally finite kernel and residually finite quotient, both MF, so the MF property is not closed under extensions; and F_8 surjects onto E , so it is not closed under quotients (Section S11, Corollary S9.8).

Supplementary material

Section S2 proves $w \neq 1$ and Section S11.3 proves E sofic; both are used by the main text and neither is a corollary of it.

S1. THE BASE, AND A SMALLER PRESENTATION

The base $\mathcal{B}$ of (7) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, for which property (T) is classical [BHV, Example 1.7.4(i)].

Six generators suffice. The group E is generated by the six elements v_1, x, y, z, t, c , and has a Tietze-equivalent presentation on them with thirty-two relators. In (8), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations; the remaining x -action relation follows from $x^3 = 1$, and the stable-letter, c -commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c .

The reduction leaves the configuration of Figure 1 unchanged: the surviving relations still give $[d, \iota(\mathcal{B})t^{-1}] = 1$ for $d = tct^{-1}$, and still leave $[d, \iota(v_1)d\iota(v_1)^{-1}]$ unconstrained.

S1.1. Property (T) of the base.

Proposition S1.1. *The twenty-relator group $\mathcal{B}$ has property (T), in both the real-orthogonal and complex-unitary formulations.*

Proof. By Remark S1.2 the presentation (7) defines $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) [BHV, Example 1.7.4(i)]. For an orthogonal representation ρ on a real Hilbert space $H_{\mathbb{R}}$, the complexification $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ carries the unitary representation $\rho \otimes 1$, whose invariant vectors and displacement norms are those of ρ in each of the two real coordinates; a Kazhdan pair for the unitary formulation is therefore one for the orthogonal formulation. $\square$

Remark S1.2 (the classical identification). The eight relations (6) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2], with the matrix-labelled generators also recorded in [CLV], and the base $\mathcal{B}$ is then the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$: the surjection $\mathcal{B} \rightarrow \bar{\Gamma}$ of Lemma S2.2 onto the matrix realization is an isomorphism.

S2. NONTRIVIALITY OF w

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra’s hyperlinear-profile construction [Slo18, Section 2].

Construction S2.1 (Clifford groups). For a set X , use the real Clifford-algebra convention generated by $(c_x)_{x \in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\mathrm{Cl}(X)$ be the subgroup of units generated by -1 and the c_x . Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\text{Cl}(X)$ is a central extension, by the sign ζ , of the elementary abelian 2-group on the points of X : the commutator of two distinct generators is ζ , where the generators of a classical lamplighter commute. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ .

Lemma S2.2 (linear model). *For $M \in \text{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

Let $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \text{diag}(2, 2, 2, 1)$. The six matrices satisfy all twenty relators of $\mathcal{B}$, and

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z . Moreover the matrix assigned to v_1 does not lie in $\text{range}(\bar{\alpha})$.

Proof. The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\text{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1D$ has entry $1/2$ in position $(1, 4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. $\square$

Form the direct limit $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$, the direct limit of $\bar{\alpha}$, extend it by its shift τ , and call the resulting group $\Sigma = T \rtimes \mathbb{Z}$; this is the group G of Theorem 5.1 at the data (5). Let $j: \bar{\Gamma} \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma/j(\bar{\Gamma})$, with base point $o = j(\bar{\Gamma})$. Since $D^{-1}\bar{v}_1D \notin \bar{\Gamma}$ (Lemma S2.2), we have $\tau o \neq j(\bar{v}_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition S2.3. *$w \neq 1$ in E .*

Proof. Form $W := \text{Cl}(X) \rtimes \Sigma$ as in Construction S2.1. Define a map on the generators of (8) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma S2.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o , so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma S2.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W , the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E . $\square$

When X is infinite so is $\text{Cl}(X)$, but each of its elements involves only finitely many Clifford generators, and every element of Σ used above is a rational 4×4 matrix. The two points are distinct because $D^{-1}\bar{v}_1 D$ has the entry $1/2$ in position $(1, 4)$ and therefore does not lie in $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$.

S3. THE MF EQUIVALENCES

An *approximate unitary representation* for a countable group G , a finite set $F \subseteq G$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$ is a map $V: G \rightarrow \mathcal{U}(r)$, for some positive dimension r , such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

The local formulation in Proposition 2.3 asks for such a model with $\delta = 1$ for every finite F and every $\varepsilon > 0$.

Proof of Proposition 2.3. Lemma 2.2 identifies $\prod_n \mathcal{U}(d_n)/\mathcal{N}$ with $\mathcal{U}(\mathcal{Q})$, which is the first equivalence once the dimension convention is normalized.

Dimension normalization. Given positive dimensions d_n , replace the n th sequence of unitary lifts by

$$\tilde{V}_n(g) = I_{\mathbb{C}^{d_0}} \oplus \cdots \oplus I_{\mathbb{C}^{d_{n-1}}} \oplus V_n(g)$$

on $\mathbb{C}^{d_0} \oplus \cdots \oplus \mathbb{C}^{d_n}$, of strictly increasing dimension $D_n = d_0 + \cdots + d_n$. Block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements, so for all g, h

$$\begin{aligned} \|\tilde{V}_n(g)\tilde{V}_n(h) - \tilde{V}_n(gh)\| &= \|V_n(g)V_n(h) - V_n(gh)\|, \\ \|\tilde{V}_n(g) - \tilde{V}_n(h)\| &= \|V_n(g) - V_n(h)\|. \end{aligned}$$

The defect and separation data are therefore unchanged, so the classes $[(\tilde{V}_n(g))_n]$ define a homomorphism to $\mathcal{U}(\mathcal{Q}((D_n)))$ — a different matrix quotient, so not the same class — which is injective for exactly the elements the original one separated. The converse is immediate.

Suppose first that $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$. For each ordered pair $g \neq h$ in F , injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer $N_{g,h}$ with $8 < N_{g,h} \delta_{g,h}^2$. The following estimate then applies: if $u, v \in \mathcal{U}(d)$ satisfy $\|u - v\| \geq \delta$, then $\|u^{\otimes p} - v^{\otimes p}\| \geq 1$ for some $p \leq N$ whenever $8 < N\delta^2$. Indeed, put $z = u^*v$, so that $\|u - v\| = \|1 - z\|$ and z has a spectral value $e^{2\pi i\alpha}$ with $|1 - e^{2\pi i\alpha}| = 2\sin(\pi\alpha') \geq \delta$, where $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$; since $(u^{\otimes p})^*v^{\otimes p} = z^{\otimes p}$ has $e^{2\pi ip\alpha}$ in its spectrum,

$$\|u^{\otimes p} - v^{\otimes p}\| \geq |1 - e^{2\pi ip\alpha}| = 2\sin(\pi \min(\{p\alpha\}, 1 - \{p\alpha\})),$$

which is at least 1 as soon as $\{p\alpha\}$ lies in the closed interval $[1/6, 5/6]$; the interval is taken closed because the stepping estimate below can land exactly on an endpoint, where the norm is exactly 1, and separation at least 1 is all that is required. The bound $2\sin(\pi\alpha') \geq \delta$ gives $\alpha' \geq \delta/(2\pi)$; consecutive multiples of α move $\{p\alpha\}$ by α' in a fixed direction modulo 1, which is less than the length $2/3$ of that interval, so some $p \leq \lceil 1/(6\alpha') \rceil \leq \pi/(3\delta) + 1 < N$ lands in it. The quotient homomorphism law makes all multiplicative defects for pairs in F arbitrarily small at every sufficiently large coordinate. Choose, for each $g \neq h$, a coordinate with the displayed separation and with every required defect at most $\varepsilon/N_{g,h}$. Then the p th tensor power separates the designated pair by at least 1, while the tensor-product telescoping identity bounds each new defect by $p\varepsilon/N_{g,h} \leq \varepsilon$. Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε . This gives the normalized approximate representation. If $|F| \leq 1$ there are no distinct pairs and hence no blocks, and the dimension is required to be positive; in that case take $r = 1$ and the map sending every element of G to $1 \in \mathcal{U}(1)$, which is exactly multiplicative and separates the empty set of pairs.

Conversely, suppose approximate unitary representations with separation constant 1 exist for all finite sets and tolerances. Choose models along an exhaustion with errors tending to zero. Each fixed pair g, h is then eventually represented multiplicatively, and each fixed pair $g \neq h$ is eventually separated by 1. Hence the classes define an injective homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$, with no condition on the dimension sequence. In particular every nonidentity element is separated by some homomorphism to a matrix quotient, so the MF residual of Definition S9.1 is trivial, and the block-sum diagonalization in the proof of Proposition S9.3(2) then assembles the separating sequences into a single faithful homomorphism to $\mathcal{U}(\mathcal{Q})$; that diagonalization uses only the dimension normalization above.

□

S4. WEIGHTS AND INTERTWINERS

Theorem 3.4 holds with an arbitrary weight ν_n in place of the dimension, and between two asymptotic representations in place of one; the theorem as stated is the case $\nu_n = d_n$.

Theorem S4.1 (one-sided conjugation at an arbitrary weight, and for intertwiners). *Let Γ , H , ι , and s be as in Theorem 3.4.*

- (1) *Let (U_n) be as there, let (ν_n) be nonnegative weights, and suppose there is C with $\text{Tr}(x_n^* x_n) \leq C \nu_n$ for all n , and that for every $\gamma \in \Gamma$ and every $\varepsilon > 0$, eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon \nu_n.$$

Then $U_n(s) x_n U_n(s)^$ and $U_n(s)^* x_n U_n(s)$ satisfy both conditions as well.*

- (2) *Let $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$ and $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$ be operator-norm asymptotic representations and let (x_n) be uniformly operator-norm bounded $d_n^1 \times d_n^2$ matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

Then $U_n^1(s) x_n U_n^2(s)^$ satisfies the same convergence for every $\gamma \in \Gamma$.*

Proof. (1) Give each coordinate space with $\nu_n > 0$ the inner product $\langle x, y \rangle = \text{Tr}(y^* x) / \nu_n$; an index with $\nu_n = 0$ forces $x_n = 0$. Conjugation by a unitary is unitary for every weight, and both hypotheses and both conclusions concern only the renormalized norms, so the ultraproduct proof of Theorem 3.4 applies with tr_{d_n} replaced by $\text{Tr}(\cdot) / \nu_n$ and nothing else changed: the bound $\text{Tr}(x_n^* x_n) \leq C \nu_n$ defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative because the weight does not enter the operator norm, and finiteness of the norm ultraproduct turns $P \leq V P V^*$ into $P = V P V^*$, which gives both directions.

(2) The block sum $U_n^\oplus = U_n^1 \oplus U_n^2$ is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. The off-diagonal embedding of x_n has operator norm $\|x_n\|$, its commutator with $U_n^\oplus(\iota(\gamma))$ is the off-diagonal embedding of the intertwining defect — whose normalized Hilbert–Schmidt norm on the block space has the denominator $d_n^1 + d_n^2$ — and conjugation by $U_n^\oplus(s)$ acts on the embedded corner by $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$. Theorem 3.4 applied to $U_n^\oplus$ gives the conclusion for the embedded sequence. Replacing the denominator $d_n^1 + d_n^2$ by an arbitrary weight ν_n , and applying (1) to the block sum in place of Theorem 3.4, gives (2) at every weight. $\square$

Taking $\nu_n = d_n$ gives Theorem 3.4 with the uniform operator-norm bound relaxed to a uniform bound on $\text{Tr}(x_n^* x_n) / d_n$. Taking $\nu_n = k_n$, the rank of the projection lift in the proof of Theorem S10.1, gives the normalization used there.

Invariant tensors of type (p, q) may be identified with Hilbert–Schmidt intertwiners from the q th to the p th tensor power. Tensor powers of unitaries are exactly multiplicative and satisfy $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$, so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem S4.1(2) applies to every pair of them: at each fixed tensor type and each weight, a sequence asymptotically invariant for the conjugate copy of the Kazhdan image is already asymptotically invariant for the whole of it.

S5. OBSTRUCTIONS FROM A ONE-SIDED CONJUGATION DATUM

In each result below a subgroup of H lies in the kernel of every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$, because it is at once 2-norm trivial and 2-norm far from the identity on a corner. The hypothesis producing the corner varies:

- Theorem S5.3: a *finite* normal subgroup of N_{conj} , the corner being the complement of its rounded averaging operator — the case of the central involution of Section 4;
- Theorem S5.6: a normal property-(T) subgroup of N_{conj} , which may be infinite, noncentral and torsion-free, the corner being a spectral projection of a Kazhdan average;
- Theorem S5.7: the same statement with the conjugation datum discarded and its only consequence, 2-norm triviality, taken as the hypothesis;
- Corollary S5.8: that statement applied to the subgroup $\mathfrak{D}(H, L)$ of (9), which depends on L alone and not on a choice of t and c .

Definition S5.1 (Kazhdan conjugation datum). Let H be a countable group. A *Kazhdan conjugation datum* in H is a tuple (Λ, ι, t, c) , where Λ is a countable group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism, not assumed injective, and $t, c \in H$, subject to

- (M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
- (M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The datum determines the normal closure

$$N_{\text{conj}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

For a finitely generated Λ , conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. The relations make c commute with $\iota(\Lambda)$ and d commute with the subgroup $t\iota(\Lambda)t^{-1}$, but they do not constrain the commutators $[d, \iota(\lambda)]$; N_{conj} is their normal closure.

Definition S5.2 (2-norm triviality). Let $U = (U_n)$ be an operator-norm asymptotic representation of a countable group H and let ω be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) = \{g \in H : \lim_\omega \|U_n(g) - 1\|_2 = 0\},$$

the kernel of the homomorphism induced by U into the tracial ultraproduct of the matrix blocks along ω , and hence a normal subgroup of H . An element g is *2-norm trivial* if $g \in K_2^\omega(U)$ for every such U and every such ω . Convergence of the full sequence, $\|U_n(g) - 1\|_2 \rightarrow 0$, gives membership along every ω at once.

Theorem S5.3 (finite-normal obstruction criterion). *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{conj}}$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Proof. *2-norm triviality of N_{conj} .* Let $U_n: H \rightarrow \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H — in particular, coordinate unitary lifts of a homomorphism to $\mathcal{U}(\mathcal{Q})$ — write $U_{g,n} = U_n(g)$, and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, its lifts belong to the asymptotic commutant in normalized Hilbert–Schmidt norm of $\iota(\Lambda)$. By Theorem 3.4 applied to $x_n = U_{c,n}$, the lifts of $d = tct^{-1}$ lie there as well. Therefore $[d, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$; since K_2 is normal, it contains the normal closure of these commutators: $N_{\text{conj}} \subseteq K_2$, and so $F \subseteq K_2$. Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of N_{conj} is 2-norm trivial (Definition S5.2).

The coordinate averaging corner. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F ; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent, and normality of F makes it asymptotically central for all of $\Theta(H)$. The Hermitian part of p_n is asymptotically idempotent, so its spectrum clusters at $\{0, 1\}$ and rounding at $1/2$ by continuous functional calculus yields genuine projections at operator-norm distance $o(1)$ from p_n ; the rounded projections are asymptotically central as well. Let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates and relabel them by $\mathbb{N}$. Compress every lift to the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with $r_n = \text{rank } q_n > 0$, and polar-correct the compressed matrices: asymptotic centrality makes each compression $q_n V_{g,n} q_n$ almost unitary in the corner, so its polar unitary is at operator-norm distance $o(1)$ and almost-multiplicativity is preserved by the correction. The result is an operator-norm asymptotic representation $(W_{g,n})$ of H on the nonzero corner blocks. Since the corner is

the complement of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Since $F \subseteq N_{\text{conj}}$ and $(W_{g,n})$ is itself an operator-norm asymptotic representation of H , the first step applies to it: every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \rightarrow 0$. Since F is finite, the maximum of these $|F|$ null sequences is itself null, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no homomorphism to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F . $\square$

The same conclusion holds with N_{conj} replaced by any normal subgroup all of whose elements lie in $K_2^\omega(U)$ for every operator-norm asymptotic representation U of H and every ω ; Theorem S5.5 is that statement for $\mathfrak{D}(H, L)$.

The central involution obstruction (Theorem 4.1) is the case $F = \{1, w\}$, where $p_n = \frac{1}{2}(1 + V_{w,n})$ and the complementary corner gives a model in which $w = -1$ exactly.

S5.1. An intrinsic form. For a subgroup $L \leq H$, let $G^+(L)$ be generated by the elements s satisfying $sLs^{-1} \subseteq L$ — the *one-sided conjugators* of L — and put

$$(9) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G^+(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H,$$

which depends only on the pair (H, L) , not on a choice of datum.

Corollary S5.4 (every one-sided conjugator acts, in both directions). *Let (r_n) and $V_{g,n} \in \mathcal{U}(r_n)$ be an operator-norm asymptotic representation of H , let Λ have property (T), let $\iota: \Lambda \rightarrow H$ be a homomorphism, and put $L = \iota(\Lambda)$. Suppose that $(x_n \in M_{r_n}(\mathbb{C}))$ is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of L :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

Then every $g \in G^+(L)$ acts in both directions on that asymptotic commutant. Explicitly, both

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$.

Proof. For a one-sided conjugator s , Theorem 3.4 gives the conclusion for $V_{g,n} x_n V_{g,n}^*$, and the equality $P = VPV^*$ established in its proof gives $V^*PV = P$, hence $V^*\text{Fix} = \text{Fix}$ and the conclusion for $V_{g,n}^* x_n V_{g,n}$. The elements acting in both directions form a subgroup of H , which therefore contains $G^+(L)$. $\square$

Theorem S5.5 (the subgroup $\mathfrak{D}$ and the MF residual). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \leq \mathfrak{D}(H, L)$, then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps every element of F to the identity.*

Proof. Let $(U_{h,n})$ be coordinate lifts of an operator-norm asymptotic representation of H and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$, a normal subgroup of H . If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant of L , and by Corollary S5.4 so does $(U_{gzg^{-1},n})$ for every $g \in G^+(L)$. Hence each generator $[gzg^{-1}, \ell]$, $\ell \in L$, displayed in (9) belongs to K_2 ; since $K_2 \trianglelefteq H$, it contains their normal closure: $\mathfrak{D}(H, L) \leq K_2$. Every element of $\mathfrak{D}(H, L)$ is therefore 2-norm trivial in every operator-norm asymptotic representation of H . The proof of Theorem S5.3 uses only that triviality, so it applies to any finite normal $F \leq \mathfrak{D}(H, L)$. $\square$

S5.2. The normal-Kazhdan form of the obstruction. A finite group has property (T), and its averaging operator is the projection onto its fixed space. The spectral projection of a Kazhdan average replaces that operator for an arbitrary normal property-(T) subgroup, which may be infinite, noncentral and torsion-free.

Theorem S5.6 (normal-Kazhdan obstruction). *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c) , and let $K \trianglelefteq H$ be a normal subgroup with property (T) such that $K \subseteq N_{\text{conj}}$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Finite groups have property (T), so $K = F$ recovers Theorem S5.3. In Theorem S5.6, K may be infinite, noncentral, and torsion-free.

Proof. The containment $N_{\text{conj}} \subseteq K_2$ from the proof of Theorem S5.3 holds for any normal subgroup, so every element of K is 2-norm trivial in every operator-norm asymptotic representation of H .

Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K . Its image $\bar{H} = \Theta(H)$ is countable, hence MF, and Proposition 2.3, applied along an exhaustion of $\bar{H}$, gives an operator-norm asymptotic representation $(V_{g,n})$ in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models bound only operator-norm distances; at growing dimension they give no lower bound on normalized Hilbert–Schmidt distances. The datum and the subgroup pass to their Θ -images with normality and property (T) preserved, and $\bar{K} = \Theta(K)$ is nontrivial and lies in the subgroup N_{conj} determined by the image datum; it suffices to derive a contradiction for $\bar{H}$.

The Kazhdan corner. Fix a free ultrafilter ω , let H_ω be the Hilbert-space ultraproduct of the coordinate spaces $\mathbb{C}^{d_n}$, and let $B_\omega = \prod_\omega B(\mathbb{C}^{d_n})$ act on it as in Section 3, so that the classes $\pi(g) = [V_{g,n}]_\omega$ define a unitary representation of $\bar{H}$ on H_ω . Fix a finite symmetric generating Kazhdan set

$S \subseteq \bar{K}$ with constant ε_0 , put $\theta = 1 - \varepsilon_0^2/(4|S|)$, and let

$$h = \frac{1}{|S|} \sum_{a \in S} \pi(a) \in B_\omega,$$

self-adjoint because S is symmetric. A unit vector ξ has $\langle h\xi, \xi \rangle = 1$ exactly when $\pi(a)\xi = \xi$ for every $a \in S$, while the Kazhdan pair bounds $\langle h\xi, \xi \rangle$ by θ on the orthogonal complement of $\text{Fix } \pi(\bar{K})$. The spectrum of h therefore lies in $[-1, \theta] \cup \{1\}$, the spectral projection $P = \chi_{\{1\}}(h)$ belongs to B_ω , and its range is $\text{Fix } \pi(\bar{K})$. Put $q = 1 - P$.

If $q = 0$ then $\pi(k) = 1$ for every $k \in \bar{K}$, against the operator-norm separation of a fixed nontrivial element of $\bar{K}$. For $g \in \bar{H}$ the projection $\pi(g)P\pi(g)^*$ has range $\text{Fix } \pi(g\bar{K}g^{-1}) = \text{Fix } \pi(\bar{K})$ by normality, so P and q commute with $\pi(\bar{H})$ exactly. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero along ω and with $\|[q_n, V_{g,n}]_-\| \rightarrow_\omega 0$ for every $g \in \bar{H}$.

Convergence along ω is not convergence as $n \rightarrow \infty$, so pass to a subsequence before compressing. Enumerate $\bar{H} = \{g_1, g_2, \dots\}$. For each j the set

$$A_j = \{n : q_n \neq 0 \text{ and } \|[q_n, V_{g_i,n}]_-\| < 1/j \text{ for all } i \leq j\}$$

is a finite intersection of sets in ω , hence lies in ω and in particular is infinite. Choose $n_1 < n_2 < \dots$ with $n_j \in A_j$ and relabel along $(n_j)_j$. On the relabelled sequence $q_n \neq 0$ for every n and $\|[q_n, V_{g,n}]_-\| \rightarrow 0$ in the ordinary sense for every fixed $g \in \bar{H}$, since $g = g_i$ lies in the constraint of A_j for all $j \geq i$. Compressing every model to $q_n M_{d_n}(\mathbb{C}) q_n$ and polar-correcting now yields an operator-norm asymptotic representation $(W_{g,n})$ of $\bar{H}$ on nonzero blocks.

A 2-norm separation on the corner. Equip each corner block with its renormalized trace. The corner compression satisfies $ghq \leq \theta q$, so at ω -most coordinates the average over S of the real parts of the renormalized traces of the compressed models is at most $\theta + o(1)$. Some generator in S therefore has compressed model with real renormalized trace at most $\theta + o(1)$, and so normalized Hilbert–Schmidt distance at least $\sqrt{2(1-\theta)} - o(1)$ from the corner identity; passing to a subsequence fixes a single such generator s_0 .

The contradiction. The corner models form an operator-norm asymptotic representation of $\bar{H}$, and $\bar{K}$ lies in the subgroup N_{conj} determined by the image datum, so 2-norm triviality applies to them: every element of $\bar{K}$ — in particular s_0 — satisfies $\|W_{s_0,n} - 1\|_2 \rightarrow 0$ on the corner, contradicting the separation just obtained. $\square$

Theorem S5.7 (obstruction from 2-norm triviality). *Let H be a countable group and let $D \leq H$ be a subgroup every element of which is 2-norm trivial (Definition S5.2). Let $K \trianglelefteq H$ be a normal subgroup with property (T) such that $K \leq D$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Proof. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K . The image $\bar{H} = \Theta(H)$ is countable and MF, and Proposition 2.3 provides an

operator-norm asymptotic representation of $\bar{H}$ separating every fixed nontrivial element. Every operator-norm asymptotic representation of $\bar{H}$ composes with Θ to one of H , so every element of $\Theta(D)$ is again 2-norm trivial, while $\bar{K} = \Theta(K) \leq \Theta(D)$ is normal, Kazhdan and nontrivial. The spectral projection q_n , the 2-norm separation at some $s_0 \in \bar{K}$, and the compression constructed in the proof of Theorem S5.6 use only that $\bar{K}$ is normal, Kazhdan and nontrivial, and 2-norm triviality of s_0 on the corner contradicts that separation. $\square$

Applied to the intrinsic defect, Theorem S5.7 gives the companion of Theorem S5.5 for a normal property-(T) subgroup:

Corollary S5.8 (normal-Kazhdan form of the intrinsic residual). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $K \trianglelefteq H$ is a normal subgroup with property (T) and $K \leq \mathfrak{D}(H, L)$, then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps every element of K to the identity.*

Proof. By the proof of Theorem S5.5, every element of $\mathfrak{D}(H, L)$ is 2-norm trivial in every operator-norm asymptotic representation of H . Theorem S5.7 with $D = \mathfrak{D}(H, L)$ concludes. $\square$

The subgroup generated by *all* normal property-(T) subgroups of $\mathfrak{D}(H, L)$ therefore also lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$.

S6. THE FINITE-DIMENSIONAL OBSTRUCTION

Every finite-dimensional representation of E over every field maps w to the identity (Theorem 3.3 with $H = E$). So finite-dimensional linear representations do not separate the points of E , the image of w is trivial in every finite quotient, and the Clifford quotient of Section S2 is infinite. The central involution w lies in the kernel of every homomorphism to a finite group, so E is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E is such a counterexample for the operator norm.

The same calculation proves the final, subgroup-valued assertion of Theorem 3.3. The elements of H whose conjugation action preserves C in both directions form a subgroup. Every one-sided conjugator lies in it by the argument above, hence so does all of $G^+(\Gamma)$. Thus, for $g \in G^+(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence every displayed generator in (9) lies in $\ker \pi$; since $\ker \pi \trianglelefteq H$, the whole normal closure $\mathfrak{D}(H, \Gamma)$ does as well.

Corollary S6.1. *The group E of Definition 6.1 admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.*

Proof. Apply Theorem 3.3 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1 ; the hypotheses hold by the defining relations (8). Every finite-dimensional representation maps w to the identity, and $w \neq 1$ by Proposition S2.3. For a finite target, compose with its faithful left regular representation on the group algebra over $\mathbb{Q}$. The composite is a finite-dimensional linear representation, so it maps w to the identity. Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of w in the finite group. $\square$

S7. THE CYCLIC COMPARISON

Theorem S7.1 (the cyclic comparison). *Define the presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_{BS} is finitely presented and $w_{\text{BS}} \neq 1$. There is an explicit surjective homomorphism from E_{BS} onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps w_{BS} to the nontrivial central sign. Nevertheless every finite-dimensional representation of E_{BS} over every field maps w_{BS} to the identity.

So a distinguished word can be trivial in every exact finite-dimensional representation without being trivial in the group.

Proof of Theorem S7.1. As with w , the relator $w_{\text{BS}}^2 = 1$ is redundant: tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (4). Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. By the commutant argument of Theorem 3.3, every finite-dimensional representation over every field maps w_{BS} to the identity.

For nontriviality, use the affine realization of $\text{BS}(1, 2)$ and its coset action. The base point is fixed by the cyclic base, while the two points represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these points and map c to the generator at the base point. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\text{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient. $\square$

Sharpness of the Kazhdan hypothesis. The affine group $\text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is solvable and the Clifford group on its coset space is locally finite, so their semidirect product is amenable. The Clifford quotient in Theorem S7.1 is a subgroup of it, hence amenable and, by [TWW], MF; its distinguished word is the nontrivial Clifford sign by the calculation above. Composing the quotient map with a faithful homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $\Theta: E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta(w_{\text{BS}}) \neq 1$, so property (T) cannot be omitted from the central involution obstruction; in the residual language of Section S9, $w_{\text{BS}} \notin \text{Res}_{\text{MF}}(E_{\text{BS}})$.

The homomorphism to $\mathcal{U}(\mathcal{Q})$ just constructed cannot be assembled from exact finite-dimensional representations. In every finite group the doubling relation makes the image of $\langle \gamma_0^2 \rangle$ equal to the image of $\langle \gamma_0 \rangle$, so the generator commutes with the required “halved” element and the distinguished word maps to the identity. A defining relator is already the identity *as an element* of the presented group, so containing it in the local set says nothing; what is needed is that the partial multiplication table evaluate the defining words and the distinguished word correctly. Take, then, a finite set containing the generators together with every intermediate prefix of every defining word and of the distinguished word. An exact local embedding of such a set satisfies the same equations. Hence neither E_{BS} nor its Clifford quotient is LEF, and the latter is not residually finite.

S8. THE SCALING FAMILY

For $m \geq 2$, let E_m be the same eight-generator presentation, replacing the three relations $tv_it^{-1} = v_i^2$ by $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged; thus $E_2 = E$. Write w_m for the image of the same distinguished word w .

Corollary S8.1 (the scaling family). *For every natural number $m \geq 2$, the group E_m is finitely presented and w_m is a nontrivial central involution mapped to the identity by every homomorphism $E_m \rightarrow \mathcal{U}(\mathcal{Q})$. Consequently E_m is not MF, and neither $C_{\max}^*(E_m)$ nor $C_r^*(E_m)$ is an MF C^* -algebra.*

Proof. Replace the affine dilation $\text{diag}(2, 2, 2, 1)$ by $\text{diag}(m, m, m, 1)$ (both acting as in Lemma S2.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Its inverse sends v_1 to translation by e_1/m , which is not integral; hence the two Clifford generators remain distinct and w_m maps to -1 . The Kazhdan base, the one-sided conjugation argument, and the central involution obstruction are independent of m . $\square$

S9. THE MF RESIDUAL

Definition S9.1. For a countable group G_1 , the *MF residual* is

$$\text{Res}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a homomorphism } G_1 \rightarrow \mathcal{U}(\mathcal{Q}) \},$$

a normal subgroup of G_1 . If G_1/N is MF then composing $G_1 \rightarrow G_1/N$ with an injective homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ gives a homomorphism with kernel N , so $\text{Res}_{\text{MF}}(G_1) \leq N$: it is the smallest normal subgroup with MF quotient, which is the sense in which *residual* is used here and for the nilpotent and finite residuals. The intersection is the same whether the dimension sequences are required to be strictly increasing or allowed to be arbitrary positive (Proposition 2.3). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

Lemma S9.2 (functoriality). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_2).$$

In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \text{Res}_{\text{MF}}(G_1)$ and Θ is a homomorphism $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$, then $\Theta \circ \varphi$ is a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$, so $\Theta(\varphi(x)) = 1$. For the second statement, $w \in \text{Res}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Res}_{\text{MF}}(G_2)$, and an injective homomorphism $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$ would map $\varphi(w)$ to 1, a contradiction. $\square$

Proposition S9.3 (the universal MF quotient). *Let G_1 be a countable group and $R = \text{Res}_{\text{MF}}(G_1)$.*

- (1) *R is the kernel of a single homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$.*
- (2) *G_1/R is MF.*

(3) Every homomorphism from G_1 to an MF group factors through G_1/R . In particular G_1 is MF if and only if $R = \{1\}$.

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ is an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, then $j \circ \varphi$ is a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$, so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). Work on $Q = G_1/R$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ maps R to 1, and for each $g \notin R$ there is one with $\Theta(g) \neq 1$, which descends; so for every $\bar{g} \neq 1$ in Q there is a homomorphism $\Theta_{\bar{g}}: Q \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta_{\bar{g}}(\bar{g}) \neq 1$. Enumerate the nonidentity elements $\bar{g}_1, \bar{g}_2, \dots$ and these representations, and pass to coordinate unitary lifts. At stage k , choose in the i th lift, for each $i \leq k$, a coordinate at which multiplicativity holds within $1/k$ on the first k group elements and at which *its own* designated element $\bar{g}_i$ stays separated by its fixed positive constant; such a coordinate exists because $\Theta_{\bar{g}_i}$ separates $\bar{g}_i$ along infinitely many coordinates, which is all the hypothesis gives. Take the block sum of the k chosen coordinates. Block operator norms are maxima, so the single block coming from $\Theta_{\bar{g}_i}$ already separates $\bar{g}_i$, and the sum therefore separates all of $\bar{g}_1, \dots, \bar{g}_k$ at once. The resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful homomorphism $Q \rightarrow \mathcal{U}(\mathcal{Q})$. This proves (2), and composing with $G_1 \twoheadrightarrow Q$ proves (1). Finally, if G_1 is MF then an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $R = \{1\}$; if $R = \{1\}$ then (2) says G_1 is MF. $\square$

Corollary S9.4 (exact residual from a candidate quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Res}_{\text{MF}}(G_1)$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ factors uniquely through G_1/N . If in addition G_1/N is MF, then*

$$\text{Res}_{\text{MF}}(G_1) = N.$$

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every such Θ . For the reverse containment in the displayed equality, compose the quotient map $G_1 \rightarrow G_1/N$ with an injective homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$: its kernel is exactly N , while the MF residual lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$. $\square$

Corollary S9.5 (residual reduction to the quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Res}_{\text{MF}}(G_1)$ and let $q: G_1 \rightarrow G_1/N$ be the quotient map. Then*

$$\text{Res}_{\text{MF}}(G_1) = q^{-1}(\text{Res}_{\text{MF}}(G_1/N)).$$

Proof. For the forward containment, Lemma S9.2 gives $q(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_1/N)$. Conversely, suppose $q(x) \in \text{Res}_{\text{MF}}(G_1/N)$ and let Θ be a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ maps N to 1 by hypothesis, so Θ factors as $\bar{\Theta} \circ q$ with $\bar{\Theta}$ a homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$; hence $\Theta(x) = \bar{\Theta}(q(x)) = 1$. $\square$

Corollary S9.6. *Let A be a unital C^* -algebra admitting an injective, possibly nonunital, $*$ -homomorphism into a matrix quotient. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

Proof. Let $j: A \rightarrow \mathcal{Q}$ be the possibly nonunital inclusion into the ambient matrix quotient and let $\phi: E \rightarrow \mathcal{U}(A)$. Complement correction, as in the proof of Theorem A, gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\hat{j} \circ \phi$ is a homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. By Theorem A, $(\hat{j} \circ \phi)(w) = 1$; injectivity of $\hat{j}$ gives $\phi(w) = 1$. Proposition S2.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

S9.1. Permanence and its failure.

Lemma S9.7 (permanence). *Let all groups be countable.*

- (1) *Every subgroup of an MF group is MF.*
- (2) *Every residually finite group is MF.*
- (3) *Every locally finite group is MF.*

Proof. (1) Restrict an injective homomorphism to $\mathcal{U}(\mathcal{Q})$. (2) For a finite test set F , a finite quotient injective on F , composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 2.3 concludes. (3) A finite test set F lies in a finite subgroup K . The left regular representation $\lambda_K: K \rightarrow \mathcal{U}(\mathbb{C}^K)$ is a finite-dimensional unitary representation of K by permutation matrices. A local model is a map defined on all of G , so set

$$V(g) = \lambda_K(g) \quad (g \in K), \quad V(g) = 1 \quad (g \notin K).$$

On the test set V is exactly multiplicative, since a product of test elements again lies in K and λ_K is a homomorphism there; the values outside K are never used in the two conditions, which quantify over F only. It separates as in (2). $\square$

Corollary S9.8 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and the explicit group E is its quotient but is not MF.*

Proof. Map the eight free generators onto the eight displayed generators of E . The resulting homomorphism is surjective because those generators define the displayed presentation. The group F_8 is residually finite, hence MF by Lemma S9.7(2), while E is not MF by Theorem A. $\square$

S10. ORBIT COLLAPSE

The argument of Theorem 4.1 extends from the negative spectral corner of a central involution to an arbitrary projection of $\mathcal{Q}$ with commuting conjugation orbit.

Theorem S10.1 (a commuting orbit forces commutation). *Let H be countable, let $L \leq H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism. Let $p \in \mathcal{Q}$ be a projection such that p commutes with $\Theta(s\gamma s^{-1})$ for every $\gamma \in L$ and the projections $\Theta(\gamma)p\Theta(\gamma)^*$, $\gamma \in L$, commute pairwise. Then p commutes with $\Theta(L)$.*

Proof. Property (T) makes L finitely generated; fix a finite symmetric generating set $S = \{a_1, \dots, a_m\}$, so that $S = S^{-1}$ and every element of L is a product of elements of S with no inverses taken. Write $p_g = \Theta(g)p\Theta(g)^*$ and $d_g = p_g - p$ for $g \in L$, and suppose $d_{g_0} \neq 0$ for some $g_0 \in L$.

Since p_g and p are commuting projections, d_g is self-adjoint with $d_g^3 = d_g$, so d_g^2 is a projection and $d_g d_g^2 = d_g$; write $q_g = d_g^2$. All the p_g commute, hence so do all the d_g and all the q_g . Put

$$q = 1 - \prod_{a \in S} (1 - q_a),$$

the join of the projections q_a , $a \in S$. The elements g with $p_g = p$ form a subgroup of L , so if $q = 0$, and hence $d_a = 0$ for every $a \in S$, then $d_g = 0$ for every $g \in L$, against $d_{g_0} \neq 0$. Thus $q \neq 0$.

A Hilbert space attached to q . Take a self-adjoint bounded lift (x_n) of q . Then $\|x_n^2 - x_n\| \rightarrow 0$, so the spectrum of x_n clusters at $\{0, 1\}$ and continuous functional calculus makes $Q_n = \chi_{[1/2, \infty)}(x_n)$ a projection with $\|Q_n - x_n\| \rightarrow 0$; the sequence (Q_n) again lifts q . Since $q \neq 0$, $Q_n \neq 0$ for infinitely many n ; discard the remaining coordinates, relabel by $\mathbb{N}$, and put $k_n = \text{rank } Q_n \geq 1$.

Give $M_{d_n}(\mathbb{C})$ the inner product $\langle x, y \rangle_n = \text{Tr}(y^*x)/k_n$, write $\|x\|_F = \text{Tr}(x^*x)^{1/2}$ for the unnormalized Hilbert–Schmidt norm, and let K_ω be the Hilbert-space ultraproduct of these spaces along a free ultrafilter ω . Let $\mathcal{I}_q$ be the set of finite sums $\sum_{j \leq r} x_j q y_j$ with $x_j, y_j \in \mathcal{Q}$, a two-sided ideal of $\mathcal{Q}$ invariant under inner automorphisms.

Given such a sum, choose bounded coordinate lifts $X_{j,n}$ of x_j and $Y_{j,n}$ of y_j and put $Z_n = \sum_{j \leq r} X_{j,n} Q_n Y_{j,n}$, of rank at most $r k_n$. The square of the Frobenius norm is the sum of the squares of the singular values, of which at most $\text{rank}(Z_n)$ are nonzero and each is at most the operator norm, so

$$(10) \quad \|Z_n\|_F^2 \leq \text{rank}(Z_n) \|Z_n\|^2 \leq r k_n \|Z_n\|^2$$

and (Z_n) is a bounded sequence for the inner products $\langle \cdot, \cdot \rangle_n$. Its class in K_ω depends only on $z = \sum_j x_j q y_j$: a sequence Z'_n obtained from a second decomposition, with r' summands, satisfies $\|Z_n - Z'_n\| \rightarrow 0$ and $\text{rank}(Z_n - Z'_n) \leq (r + r')k_n$, so $\|Z_n - Z'_n\|_F^2/k_n \rightarrow 0$ by (10). Write $\Lambda(z) = [Z_n]_\omega$. The map $\Lambda: \mathcal{I}_q \rightarrow K_\omega$ is linear,

$$\|\Lambda(q)\| = \lim_\omega \left(\frac{\text{Tr } Q_n}{k_n} \right)^{1/2} = 1,$$

and $\Lambda(z) = 0$ implies $\Lambda(xzy) = 0$ for all $x, y \in \mathcal{Q}$, since bounded lifts of x and y multiply $\|Z_n\|_F$ by a bounded factor.

An exact cocycle. For $a \in S$ the projection q_a annihilates $\prod_{b \in S} (1 - q_b)$, so $q_a q = q_a$ and $d_a = d_a q_a = d_a q \in \mathcal{I}_q$. From $p_{gh} = \Theta(g) p_h \Theta(g)^*$,

$$(11) \quad d_{gh} = d_g + \Theta(g) d_h \Theta(g)^* \quad (g, h \in L),$$

and $\mathcal{I}_q$ is inner-invariant, so $d_g \in \mathcal{I}_q$ for every $g \in L$.

Choose unitary coordinate lifts $U_n(h)$ of $\Theta(h)$ (Lemma 2.2). Conjugation by a unitary preserves each $\langle \cdot, \cdot \rangle_n$, and $\|\text{Ad } u - \text{Ad } v\| \leq 2 \|u - v\|$, so $\pi(h) = [\text{Ad } U_n(h)]_\omega$ defines a unitary representation of H on K_ω , and $\pi(h)\Lambda(z) = \Lambda(\Theta(h)z\Theta(h)^*)$ on the closed invariant subspace $K_q = \overline{\Lambda(\mathcal{I}_q)}$. By (11) the map $\beta(g) = \Lambda(d_g)$ satisfies

$$\beta(gh) = \beta(g) + \pi(g)\beta(h) \quad (g, h \in L),$$

a 1-cocycle for $\pi|_L$ with values in K_q .

Suppose $\beta(a) = 0$ for every $a \in S$. Put $r_1 = 1$ and $r_i = \prod_{j < i} (1 - q_{a_j})$ for $i \geq 2$. The products $e_i = r_i q_{a_i}$ are projections with $e_i = r_i - r_{i+1}$, so they are pairwise orthogonal and $\sum_i e_i = 1 - r_{m+1} = q$. Each d_{a_i} commutes with r_i , so $e_i = (d_{a_i} r_i) d_{a_i}$ and $\Lambda(e_i) = 0$, whence $\Lambda(q) = 0$ against $\|\Lambda(q)\| = 1$. So β is not identically zero.

Property (T). By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4], β is a coboundary: there is $y \in K_q$ with $\beta(g) = y - \pi(g)y$ for every $g \in L$. The hypothesis on p gives $d_{sas^{-1}} = 0$, hence $\pi(sas^{-1})y = y$, for every $a \in L$. The proof of Theorem 3.4 identifies $\text{Fix } \pi(sLs^{-1})$ with $\text{Fix } \pi(L)$: the projection P onto $\text{Fix } \pi(L)$ and its conjugate $Q = \pi(s)P\pi(s)^*$, whose range is $\text{Fix } \pi(sLs^{-1})$, satisfy $P \leq Q$ and $P \sim Q$ in the norm ultraproduct, which is finite, so $P = Q$ by Lemma 3.1. The weight k_n in place of d_n leaves that comparison unchanged (Theorem S4.1(1)). Hence $\pi(g)y = y$ and $\beta(g) = 0$ for every $g \in L$, contradicting the previous paragraph. So $d_g = 0$, that is, p commutes with $\Theta(g)$, for every $g \in L$. $\square$

Equivalently, every projection of $\Theta(sLs^{-1})' \cap \mathcal{Q}$ whose $\Theta(L)$ -conjugates commute pairwise already lies in $\Theta(L)' \cap \mathcal{Q}$.

Definition S10.2 (involutive orbit witness). Let H be a group, $L \leq H$ a subgroup, and $s \in H$. An element $k \in H$ is an *involutive orbit witness* for (L, s) if

- (W1) $k^2 = 1$;
- (W2) $s\gamma s^{-1}$ commutes with k for every $\gamma \in L$;
- (W3) the conjugates $\gamma_1 k \gamma_1^{-1}$ and $\gamma_2 k \gamma_2^{-1}$ commute for all $\gamma_1, \gamma_2 \in L$.

The pair (L, s) determines the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

In a permutational wreath product $K^{(X)} \rtimes G$ with $L \leq G$ conjugated into itself by s and $x_0 \in X$ a point fixed by sLs^{-1} , every involution of the copy of K at x_0 is a witness: copies at distinct points commute, and the conjugate copy fixes the point.

Theorem S10.3 (the orbit of an involutive witness). *Let H be countable, let $L \leq H$ be a subgroup with property (T), and let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$. Then*

$$D_{\text{coll}}(L, s) \leq \text{Res}_{\text{MF}}(H).$$

Proof. Let $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism and k a witness for (L, s) . By (W1) the unitary $u = \Theta(k)$ is self-adjoint, so $p = \frac{1}{2}(1 - u)$ is a projection, and

$$\Theta(\gamma)p\Theta(\gamma)^* = \frac{1}{2}(1 - \Theta(\gamma k \gamma^{-1})) \quad (\gamma \in L).$$

By (W2) the projection p commutes with $\Theta(s\gamma s^{-1})$ for $\gamma \in L$, and by (W3) the displayed projections commute pairwise. Theorem S10.1 makes p , and hence $u = 1 - 2p$, commute with $\Theta(L)$, so $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. Kernels are normal, so Θ is trivial on $D_{\text{coll}}(L, s)$. $\square$

Theorem S10.4 (the orbit of a finite-order witness). *Let H be countable, let $L \leq H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $k \in H$ have finite order. If the conjugate copy sLs^{-1} centralizes k and the conjugates $\gamma k \gamma^{-1}$, $\gamma \in L$, commute pairwise, then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$; the normal closure of all such commutators, over all finite-order witnesses for (L, s) , lies in $\text{Res}_{\text{MF}}(H)$, with equality whenever the corresponding quotient of H is MF.*

Proof. Let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$, m the order of k , and $u = \Theta(k)$, a unitary with $u^m = 1$. Its discrete Fourier idempotents $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t / m} u^t$, for $j \in \mathbf{Z}/m$, are self-adjoint idempotents of $\mathcal{Q}$. Each p_j is a polynomial in u , so whatever commutes with u commutes with every p_j : the represented conjugate copy centralizes each p_j , and the $\Theta(L)$ -conjugates of each p_j commute pairwise because the conjugated witnesses do. Theorem S10.1 therefore makes every p_j commute with $\Theta(L)$, and Fourier reconstruction expresses $u = \sum_j e^{2\pi i j / m} p_j$, so u commutes with $\Theta(L)$ and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. $\square$

Proof of Theorem B. Write $q: E \rightarrow E/\langle w \rangle$, $\bar{d} = q(d)$, and $\bar{L} = q(\iota(\mathcal{B}))$. The element $\bar{d}$ is an involution because $c^2 = 1$, and $q(t)\bar{L}q(t)^{-1}$ centralizes $\bar{d}$ because $t\iota(\mathcal{B})t^{-1}$ commutes with c in E . Property (T) passes to $\bar{L}$, and $q(t)$ conjugates it into itself. Conditions (W1) and (W2) of Definition S10.2 therefore hold. It remains to verify (W3).

The linear generators x, y, z commute with d , and t carries each v_i to v_i^2 , so $\iota(v_i^2)$ commutes with d as well. Hence $\iota(g)d\iota(g)^{-1}$ depends only on the exponent vector of the translation part of g modulo 2, which leaves eight conjugates, indexed by $(\mathbb{Z}/2)^3$. By Theorem 5.1 the distinguished word is $w = [d, \iota(v_1)d\iota(v_1)^{-1}]$, the commutator attached to the class of v_1 . The six words x, x^2, z, yz, yx, y^2x in the linear generators carry that class onto the six remaining nonzero classes, and each commutes with d ; conjugating the displayed identity by the corresponding $\iota(r)$ exhibits every nonzero parity commutator as a conjugate of w , while the zero class contributes $[d, d] = 1$.

Since w is central, all eight lie in $\langle w \rangle$, and one further conjugation reduces an arbitrary pair of conjugates to this case. Thus $\bar{d}$ is an involutive orbit witness for $(\bar{L}, q(t))$, and by Theorem S10.3 $[q(\iota(v_1)), \bar{d}] \in \text{Res}_{\text{MF}}(E/\langle w \rangle)$.

The homomorphism $E \rightarrow W$ of Proposition S2.3 sends $[\iota(v_1), d]$ to the product of the Clifford generators at the two distinct points τo and $j(\bar{v}_1)\tau o$, and sends $\langle w \rangle$ into $\langle \zeta \rangle$, whose configuration coordinate is zero. Hence $[\iota(v_1), d] \notin \langle w \rangle$, so its image in $E/\langle w \rangle$ is nontrivial; a nontrivial element of the MF residual makes the quotient non-MF by Proposition S9.3. Corollary S9.5 identifies $\text{Res}_{\text{MF}}(E)$ with the q -preimage of $\text{Res}_{\text{MF}}(E/\langle w \rangle)$, so u belongs to it. $\square$

Corollary S10.5 (collapse computes the residual). *In the setting of Theorem S10.3, write $q: H \rightarrow H/D_{\text{coll}}(L, s)$ for the quotient map. Then*

$$\text{Res}_{\text{MF}}(H) = q^{-1}(\text{Res}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

and if $H/D_{\text{coll}}(L, s)$ is MF, then $\text{Res}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$.

Proof. Combine Theorem S10.3 with Corollaries S9.5 and S9.4. $\square$

In the second case $H/D_{\text{coll}}(L, s)$ is the universal MF quotient of H (Proposition S9.3): every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ factors through it, and it is MF.

S11. GROUP ALGEBRAS, SOFICITY, AND PERMANENCE

S11.1. The reduced group algebra of E . The stable finiteness in Theorem C comes from two standard facts.

Lemma S11.1. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_r^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.*

Proof. (1) Realize $C_r^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g . Let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$. Each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_r^*(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ implies $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

S11.2. The Clifford quotient and permanence. The map $E \rightarrow W$ of Proposition S2.3 is surjective: its image contains Σ and the generator c_o at the base point, hence every Clifford generator by transitivity of the Σ -action on X . Since E is generated by v_1, x, y, z, t, c (Section S1), their six images generate W . The map sends $w \in \text{Res}_{\text{MF}}(E)$ to the nontrivial sign ζ ; functoriality (Lemma S9.2) makes $\zeta \in \text{Res}_{\text{MF}}(W)$. Hence the finitely generated group W is not MF, and neither is $C_r^*(W)$: complement correction turns an MF embedding of $C_r^*(W)$ into an injective homomorphism $W \rightarrow \mathcal{U}(\mathcal{Q})$, as in the proof of Theorem A.

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \text{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel is locally finite — hence LEF, sofic and MF — and whose quotient is MF, while W itself is finitely generated and not MF. Nearby approximation properties do have extension permanence of this shape: an extension of a residually finite group by a residually finite group is weakly sofic [Gl], and weak soficity is detected by finite targets [GR].

Proof. By Lemma S9.7(3), the Clifford group is locally finite, hence MF. The group Σ identifies with the subgroup of $\text{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D : the levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant gives the $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the direct limit and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus Σ is MF by Lemma S9.7(2), whereas W is not MF by the preceding construction. $\square$

Extensions by $\mathbb{Z}$. Composing the two canonical projections gives a surjection $W \rightarrow \mathbb{Z}$ whose kernel is $\text{Cl}(X) \rtimes T$, and this kernel is LEF, hence MF: every finite subset lies in $L \rtimes \bar{\Gamma}_n$ for a finite invariant subgroup $L \leq \text{Cl}(X)$ and one level, and that group is residually finite because the level acts on the finite group L through a finite quotient. Thus W is an extension of an LEF group by $\mathbb{Z}$ and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06].

A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\text{Res}_{\text{MF}}(S_{\text{simp}}) = S_{\text{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed E , which is sofic by Theorem D, in such a group. Functoriality of the MF residual sends w to a nontrivial element of the MF residual of the simple group. That residual is normal, so it is the whole group. Every homomorphism to an MF group maps the MF residual to 1. $\square$

S11.3. Soficity of the literal group. Deleting the letter c and the families (C) and (W) from (8) leaves the presentation $\langle \mathcal{B}, t \mid tut^{-1} = \bar{\alpha}(u) \rangle$ of the ascending HNN extension of the base along the doubling endomorphism. By Remark S1.2 the base is $\bar{\Gamma}$, so

$$E/\langle\langle c \rangle\rangle \cong \Sigma$$

for the group $\Sigma = T \rtimes \mathbb{Z}$ of Section S2, with its site set $X = \Sigma/B$, base point o , shift τ and level-zero base $B = j(\bar{\Gamma})$. The levels

$$\bar{\Gamma}_n = \tau^{-n} B \tau^n \quad (n \geq 0)$$

form an ascending chain of copies of $\bar{\Gamma}$ with union T . Put $K = \tau^{-1} B \tau$; then $B \leq K$ and $[K : B] = [\bar{\Gamma} : \bar{\alpha}(\bar{\Gamma})]$, at most eight. The cosets Σ/K are the *blocks*, and each block, as a set of sites, has $[K : B]$ elements.

Let $\mathcal{G}$ be the graph on X whose edges are the Σ -orbit of the pair $\{\tau o, j(\bar{v}_1)\tau o\}$, whose entries are distinct by Lemma S2.2. Two sites are adjacent in $\mathcal{G}$ exactly when they are distinct and lie in a common block. Adjacent sites share a block because $(\tau o)^{-1}(j(\bar{v}_1)\tau o) = \tau^{-1}j(\bar{v}_1)\tau \in K$. Conversely, the chart $u\bar{\alpha}(\bar{\Gamma}) \mapsto \tau^{-1}j(u)\tau o$ identifies the block of o with $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ and the generating edge with the pair $\{\bar{\alpha}(\bar{\Gamma}), \bar{v}_1\bar{\alpha}(\bar{\Gamma})\}$; since $\bar{\alpha}(\bar{\Gamma})$ acts transitively on the nontrivial cosets of $\bar{\alpha}(\bar{\Gamma})$, every pair of distinct sites of a block is the image of the generating pair.

For a graph $\mathcal{G}$ on X , let $C(\mathcal{G})$ be the group presented by generators ζ and c_ξ ($\xi \in X$) and relations

$$\zeta^2 = c_\xi^2 = 1, \quad [\zeta, c_\xi] = 1, \quad [c_\xi, c_\eta] = \zeta \text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on X this is $\text{Cl}(X)$ of Construction S2.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection $C(\mathcal{G}) \rightarrow \text{Cl}(X)$.

Proposition S11.2 (normal form for E). *There is an isomorphism*

$$E \cong C(\mathcal{G}) \rtimes \Sigma$$

under which $\langle\langle c \rangle\rangle$ is the factor $C(\mathcal{G})$, the conjugate gcg^{-1} is the lamp c_{gB} , and the distinguished word w is the sign ζ . Each level $\bar{\Gamma}_n$ has finite orbits on X .

Proof. Send the six base letters and t to the corresponding generators of Σ , and c to c_o . Families (B) and (T) of (8) hold in Σ by construction. For family (C): $c_o^2 = 1$, and B fixes o , so each base generator fixes c_o . For family (W): the pair $\{\tau o, j(\bar{v}_1)\tau o\}$ is an edge, so the image of w is $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$, which is central in $C(\mathcal{G})$ and fixed by Σ . This defines $E \rightarrow C(\mathcal{G}) \rtimes \Sigma$.

In the other direction, $\zeta \mapsto w$ and $c_{gB} \mapsto gcg^{-1}$ respects the defining relations of $C(\mathcal{G})$: the assignment is independent of the coset representative because the base centralizes c ; the images are involutions; w is central of order two; and along an edge $g\{\tau o, j(\bar{v}_1)\tau o\}$ the commutator of the two

images is $gw g^{-1} = w$; the presentation imposes relations only along edges. Combined with the splitting $\Sigma \rightarrow E$ on the seven remaining letters, this gives $C(\mathcal{G}) \rtimes \Sigma \rightarrow E$. The two composites fix the eight letters of E and the generators ζ, c_ξ of $C(\mathcal{G})$, hence are the identity.

For the orbits, $\bar{\alpha}(\bar{\Gamma})$ has index at most eight in $\bar{\Gamma}$ — the parity classes of the translation lattice — so τ conjugates B onto a subgroup of finite index in itself. Every $\bar{\Gamma}_n$ is therefore commensurable with B , all of Σ commensurates B , and orbit–stabilizer makes each $\bar{\Gamma}_n$ -orbit in $X = \Sigma/B$ finite. $\square$

Composing the isomorphism with $C(\mathcal{G}) \rightarrow \text{Cl}(X)$ gives the surjection $E \rightarrow W$ of Proposition S2.3. The group $\text{Cl}(X)$ is locally finite; $C(\mathcal{G})$ is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set S of sites write

$$M_S = \langle \zeta, c_\xi \ (\xi \in S) \rangle \leq C(\mathcal{G}).$$

Lemma S11.3 (finite-site subgroups are residually finite). *For every finite set S of sites the subgroup M_S is residually finite.*

Proof. The quotient map by $\langle \zeta \rangle$ carries $C(\mathcal{G})$ onto the free product of the elementary abelian 2-groups of the blocks, and M_S into the free product of the finitely many factors at blocks meeting S . A free product of finite groups is residually finite. The kernel of $C(\mathcal{G}) \rightarrow C(\mathcal{G})/\langle\langle \zeta \rangle\rangle$ is the normal closure of ζ , and ζ is central, so that normal closure is $\langle \zeta \rangle$; hence the kernel of this map on M_S lies in $\langle \zeta \rangle$. (No splitting is available or needed: a two-lamp block is the Clifford extension $D_8 \rightarrow (\mathbb{Z}/2)^2$, which does not split over the whole factor.) The surjection $C(\mathcal{G}) \rightarrow \text{Cl}(X)$ sends ζ to the sign of $\text{Cl}(X)$, which is nontrivial, and $\text{Cl}(X)$ is locally finite, so the image of the finitely generated group M_S there is finite. An element of M_S in the kernel of the first map lies in $\langle \zeta \rangle$, and the second map separates the two elements of that subgroup. $\square$

Proof of Theorem D. Let E_0 be the kernel of the homomorphism $E \rightarrow \mathbb{Z}$ given by the exponent sum in t . By Proposition S11.2, $E_0 \cong C(\mathcal{G}) \rtimes T$.

Take a finite subset of E_0 . Its T -coordinates lie in one level $\bar{\Gamma}_n$, the chain being ascending, and its $C(\mathcal{G})$ -coordinates lie in M_{S_0} for some finite set S_0 of sites, each generator of $C(\mathcal{G})$ being a single lamp or the sign. Let S be the union of the $\bar{\Gamma}_n$ -orbits of the sites in S_0 , a finite set by Proposition S11.2. Then M_S is $\bar{\Gamma}_n$ -invariant and contains M_{S_0} , so the finite subset lies in $M_S \rtimes \bar{\Gamma}_n$.

That group is residually finite. Indeed M_S is residually finite by Lemma S11.3, the level $\bar{\Gamma}_n \cong \bar{\Gamma}$ is residually finite through the congruence quotients of its matrix model (Lemma S2.2), and $\bar{\Gamma}_n$ acts on M_S through permutations of the finite set S , hence through a finite quotient F . The preimage of the identity of F is a subgroup of finite index isomorphic to $M_S \times \ker(\bar{\Gamma}_n \rightarrow F)$, which is residually finite, and residual finiteness passes to a group containing a residually finite subgroup of finite index.

So E_0 is a directed union of subgroups embedding in residually finite groups, that is, LEF, and therefore sofic. Now E is a split extension of E_0 by $\mathbb{Z}$, split by t , so E is sofic by the Elek–Szabó amenable-extension theorem [ES06], and hyperlinear because sofic groups are. Failure of MF is Theorem A. LEF groups are MF [CDE], so E is not LEF; a residually finite group is LEF, so E is not residually finite either. $\square$

The same decomposition makes E exact. The quotient of $C(\mathcal{G})$ by $\langle \zeta \rangle$ is the free product of the elementary abelian 2-groups of the blocks, each of order at most 2^8 , and free products, directed unions and extensions of exact groups are exact [KW2]; so that free product is exact, and $C(\mathcal{G})$, a central extension of it by $\langle \zeta \rangle$, is exact as well. The group Σ is a subgroup of $\mathrm{GL}_4(\mathbb{Q})$ and therefore exact [GHW], and $E \cong C(\mathcal{G}) \rtimes \Sigma$. For a discrete group exactness is equivalent to exactness of the reduced group algebra [KW1], so $C_r^*(E)$ is exact. The same applies to W , whose Clifford kernel is locally finite and hence amenable.

S12. MARKED LIMITS, QUASI-IDENTITIES, AND UNDECIDABILITY

S12.1. Marked-limit closure.

Theorem S12.1 (the MF locus is closed). *Fix $k \geq 1$ and consider the space of k -marked groups — quotients of the free group F_k , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.*
- (2) *Equivalently, the non-MF marked groups form an open set.*
- (3) *Every non-MF k -marked group admits a finite-radius certificate: there is a radius R such that every k -marked group whose relations agree with it on the reduced words of length at most R is non-MF.*

Proof. A marked group fails MF exactly when normalized local approximation fails on some finite subset of it. Expressing the finitely many products and inequalities of that subset by words in the marked generators produces finitely many relation constraints, and every marked group satisfying those constraints has the same finite subset with no local model; so a basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form. $\square$

S12.2. A finite universal Horn obstruction. Every MF group satisfies a single finite quasi-identity: a universal implication between finitely many equations, that is, a universal Horn sentence. Write R for the forty-one relators of (8), regarded as words in eight variables, and w for the distinguished word in those variables.

Proposition S12.2 (finite quasi-identity). *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w \neq 1$.

Proof. A tuple satisfying the relators induces a homomorphism $E \rightarrow G$ sending the distinguished word of E to $w(x_1, \dots, x_8)$. If G is MF, that image is trivial: $w \in \text{Res}_{\text{MF}}(E)$ by Theorem A, and the residual is functorial (Lemma S9.2). The canonical tuple of E satisfies R by definition, and its distinguished word is nontrivial by Theorem A. $\square$

Corollary S12.3 (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group F_8 , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of R and $w \neq 1$ form a nonempty clopen set consisting entirely of non-MF groups.*

Proof. Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group E by Theorem A, and Proposition S12.2 excludes every MF point. $\square$

S12.3. Undecidability of MF recognition. Lemma S9.7(1) and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group E embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies, and classically MF recognition is undecidable.

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_e for the group presented by the code e .

Corollary S12.4 (undecidability, given the word problem). *Write W for the predicate on pairs consisting of a presentation code and a word in the generators of that code, holding when the word is trivial in the presented group. If W is undecidable, then no algorithm decides, from a presentation code, whether the presented group is MF. If the complement of W is not recursively enumerable, then neither is the set of presentation codes of non-MF groups.*

Proof. Free-product the input code with a fixed code presenting E , and apply the Adian–Rabin construction to the input word. The resulting code is computable in the pair, and presents an MF group exactly when the input word is trivial: when it is, the construction collapses the group to the trivial one; when it is not, the group contains E and Lemma S9.7(1) makes it non-MF. Composing a decision procedure for MF recognition with this map would therefore decide W . For the second assertion the same

map carries the pairs outside W exactly to codes of non-MF groups, and the preimage of a recursively enumerable set under a computable map is recursively enumerable. $\square$

Effectiveness of the coding is necessary: a map attaching a finitely presented group to each code need not be computable, and an uncomputable choice would make the statement true for reasons having nothing to do with groups.

Every group-theoretic step of that reduction is proved here, and so is the computability of the map. The Markov witness is Lemma S9.7(1) with Theorem A; correctness is the statement that the output presents an MF group exactly when the input word dies; and the transformation is computable because it is built as surgery on the relator lists rather than through free groups over a code-dependent alphabet, together with a proof that the two descriptions name the same relator set.

What is not proved here is the hypothesis, and the gap is not in the Adian–Rabin argument, which is complete above. Undecidability of W is classical, and a finitely presented group carrying a sequence of words whose triviality is undecidable is available here, with no hypothesis and no literature input, by way of Aanderaa–Cohen modular machines instead of Boone–Britton. The two have not been matched up, and the gap is coordinates rather than group theory: W is a predicate on codes, while what the existence statement supplies is a group together with a sequence of words obtained by choice. Until they are matched, Corollary S12.4 keeps its hypothesis.

The two ingredients the reduction consumes from the rest of the paper are subgroup heredity and the existence of a finitely presented non-MF group; the classical form of the conclusion, with the Markov data quantified rather than exhibited, is recorded alongside them. Finitely presented non-MF groups occur in infinite multiplicity, and finitely generated ones in continuum multiplicity. The groups $E \times \mathbb{Z}^k$ for $k \geq 0$ are finitely presented, pairwise nonisomorphic, and non-MF, each containing E (Lemma S9.7(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks; the number of homomorphisms to $\mathbb{Z}/2$ separates them as well. Beyond finite presentation, take a continuum of pairwise nonisomorphic finitely generated groups N — Neumann’s two-generator groups [Ne], say. Each product $E \times N$ is non-MF, and a fixed countable group X satisfies $X \cong E \times N$ for at most countably many of the isomorphism types N , since each such N embeds in X as a finitely generated subgroup and a countable group has only countably many of those. So the products realize $2^{\aleph_0}$ pairwise nonisomorphic finitely generated non-MF groups.

S13. LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm: the estimate

$\|\mathrm{Ad}(U) - \mathrm{Ad}(V)\| \leq 2\|U - V\|$ has no converse, and there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints are at operator-norm distance 2. The proof of Theorem 3.4 uses the operator-norm control at exactly that point: without it the classes $[\mathrm{Ad} U_n(g)]_\omega$ need not be multiplicative.

Amplification by an identity block separates the two normalizations. Padding a model with an identity block preserves every operator-norm quantity — the multiplicative defects and the pairwise separations are unchanged — while the normalized Hilbert–Schmidt distances decay with the padding ratio: below any prescribed bound, every approximate unitary representation can be replaced by one all of whose pairwise Hilbert–Schmidt distances are smaller than the bound. 2-norm triviality of an element therefore does not by itself prove a group non-MF: an obstruction must produce an invariant corner with its own normalized trace, so that the normalization is unchanged by amplification. The relative rank of the corner is unconstrained: padding leaves the averaging corner $q_n M_{d_n}(\mathbb{C}) q_n$ untouched while d_n grows, so r_n/d_n may tend to zero while the argument uses the corner’s own normalized trace. The spectral projection of Theorem S5.6 is normalized on its own range in the same way, and the proof of Theorem S10.1 normalizes by the rank k_n .

Finite dimensionality of the coordinates cannot be replaced by stable finiteness of the coordinate algebras together with faithful traces. For faithfully traced unital C^* -coordinates A , the left and right actions on $L^2(A, \tau)$ generate a von Neumann algebra that need not be finite: for an infinite-dimensional II_1 factor in standard form it is all of $B(L^2(A, \tau))$. Constant coordinates $A_n = C_r^*(E)$ give a counterexample to any such generalization: the left regular representation embeds E exactly and injectively into the norm quotient

$$\prod_n A_n / \bigoplus_n A_n,$$

the element c centralizes the Kazhdan base there, and conjugation by t produces tct^{-1} , whose commutator with the base is the element u with $u^2 = w \neq 1$ of Theorem A. Precisely: Theorem 3.4 becomes false when its matrix coordinates are replaced by stably finite unital C^* -algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]; that question is independent of Theorem 3.4. The adjoint spectral estimates used here need operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply; soficity of E comes instead from its normal form (Theorem D).

Questions.

- (1) What is $\mathrm{Res}_{\mathrm{MF}}(E)$? It properly contains $\{1, w\}$ by Theorem B, and Corollary S10.5, applied to $E/\langle w \rangle$ and its defect $D = D_{\mathrm{coll}}$, computes it from $\mathrm{Res}_{\mathrm{MF}}((E/\langle w \rangle)/D)$. Neither reduction constrains that last

group: w is already trivial in $E/\langle w \rangle$, and the commutators generating D are already trivial in the quotient by D . Some other finite-normal, normal-Kazhdan or orbit datum could still be present there.

- (2) Is there a torsion-free finitely presented non-MF group? Every finite subgroup of a torsion-free group is trivial, so Theorem S5.3 is empty there, while Theorem S5.6 applies to torsion-free groups. What remains is group-theoretic: a torsion-free finitely presented group whose subgroup N_{conj} contains a *nontrivial* normal property-(T) subgroup. The trivial subgroup is normal and has property (T) vacuously and gives no obstruction. Remark S13.1 gives a small-cancellation route over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsoficity. The Clifford construction of Theorem 5.1 cannot produce such a group: it imposes $c^2 = 1$, so $d^2 = 1$, and a torsion-free quotient q then has $q(d) = 1$ and $q(N_{\text{conj}}) = 1$.
- (3) Does MF imply hyperlinearity? A faithful embedding into a matrix quotient need not make any coordinate trace faithful, so the defining MF representation does not immediately give a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (4) For which groups do the criteria of Theorems S5.3, S5.6 and S10.1 compute the residual exactly?
- (5) Does the obstruction survive when the matrix blocks are replaced by other building blocks? The proof of Theorem 3.4 uses finite dimensionality of $M_{d_n}(\mathbb{C})$ twice: for the conjugation action on $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, and for finiteness of the norm ultraproduct. Faithful traces alone do not replace it. For which separable C^* -algebras A_n does every homomorphism $H \rightarrow \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$ still map w to the identity? The lifting of Lemma 2.1 is *not* where the matrix structure enters, and it is not a semiprojectivity question: for unital A_n and u unitary in the quotient, any lift (x_n) has $\|x_n^* x_n - 1\| \rightarrow 0$ and $\|x_n x_n^* - 1\| \rightarrow 0$, so x_n is invertible for all large n and the polar correction $u_n = x_n (x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \rightarrow 0$. Neither real rank zero nor semiprojectivity in the sense of Loring [Lo98] is used.

Remark S13.1 (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections S1–S2 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group P contains a copy of every finitely presented torsion-free group, in particular a direct product $P_1 \times P_2$ with $P_i \cong P$. Then $\alpha: P \xrightarrow{\cong} P_1 \leq P$, together with any $a \in P_2 \setminus \{1\}$, satisfies the hypotheses of Theorem 5.1. The group $E(P, \alpha, a)$ has torsion even though P does not, since the construction adjoins the involution c .

S14. THE MAXIMAL GROUP C^* -ALGEBRA

For a countable group H , the maximal group C^* -algebra $C_{\max}^*(H)$ is the completion of the group ring $\mathbb{C}[H]$ in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of H ; the supremum is finite because $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$. Write u_h for the canonical unitary of $h \in H$.

Proposition S14.1 (the universal property). *The canonical map $h \mapsto u_h$ is injective, and for every unital C^* -algebra B lying in a fixed universe that contains H , and every homomorphism $\rho: H \rightarrow \mathcal{U}(B)$, there is a unique unital $*$ -homomorphism $C_{\max}^*(H) \rightarrow B$ with $u_h \mapsto \rho(h)$ for every $h \in H$.*

Injectivity comes from the left regular representation, uniqueness from density of $\mathbb{C}[H]$. Restricting B keeps “all representations” a set, and every target below is a unitary group of a separable C^* -algebra.

Proposition S14.2 (a strictly dominated projection gives a proper isometry). *Let A be a unital C^* -algebra containing a projection p and a unitary u such that the conjugate upu^* satisfies $p(upu^*) = (upu^*)p = p$, equivalently $p \leq upu^*$, and $upu^* \neq p$. Then A contains an isometry that is not a unitary; consequently A is not stably finite and has no faithful tracial state.*

Proof. Write $q = upu^*$, so q is a projection with $pq = qp = p$, and put $r = pu^*$. Then

$$r^*r = uppu^* = upu^* = q, \quad rr^* = pu^*up = p.$$

Put $s = r + (1 - q)$. Since $p(1 - q) = 0$ and $(1 - q)p = 0$,

$$s^*s = r^*r + (1 - q) = q + (1 - q) = 1,$$

and since $pu^*(1 - q) = pu^* - pu^*upu^* = pu^* - pu^* = 0$ and likewise $(1 - q)up = 0$,

$$ss^* = rr^* + (1 - q) = p + (1 - q) = 1 - (q - p).$$

Thus s is an isometry, and $ss^* = 1$ would force $q = p$, which is excluded; so s is not a unitary. A unital C^* -algebra containing a nonunitary isometry is not finite, hence not stably finite. If τ were a faithful tracial state then $\tau(1 - ss^*) = \tau(1) - \tau(s^*s) = 0$ while $1 - ss^* = q - p$ is a nonzero positive element, contradicting faithfulness. $\square$

Remark S14.3 (the maximal algebra under proper one-sided conjugation). Let G contain a property-(T) subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$; the base of E is such a pair. Let $P \in C_{\max}^*(G)$ be the spectral projection at 1 of the Hermitian Kazhdan average $\frac{1}{2|\Gamma|} \sum_{s \in \Gamma} (u_s + u_s^*)$, which exists by the spectral gap of Section 3. Then $P \leq u_t P u_t^*$, and the inequality is strict: in the quasi-regular representation on $\ell^2(G/t\Gamma t^{-1})$ the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of Γ outside it, so

the two projections differ there. By Proposition S14.2, $C_{\max}^*(G)$ is then not stably finite, has no faithful trace, and in particular is neither residually finite-dimensional nor MF. For the reduced algebra of E , by contrast, the canonical trace is faithful and the algebra is stably finite (Theorem C), so its failure of MF is not a failure of finiteness.

ACKNOWLEDGEMENTS

We thank Caleb Eckhardt for discussions clarifying the finite-quotient analogy that organizes Section 1. We thank Francesco Fournier-Facio for formulating the group-theoretic input as a Kazhdan group that is not co-Hopfian, together with its ascending HNN extension. We also thank him for the small-cancellation construction recorded in Remark S13.1. We further thank him for helpful discussions on the distinct roles of the operator and Hilbert–Schmidt norms. We thank Alon Dogon for pointing out that the interaction between those two norms also occurs in the earlier approach of Bachner–Dogon–Lubotzky [BDL].

REFERENCES

- [AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, [arXiv:2608.05362](#).
- [BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1 -approximation of groups*, J. Algebra **702** (2026), 235–243, [doi:10.1016/j.jalgebra.2026.04.041](#).
- [Be] M. E. B. Bekka, *Amenable unitary representations of locally compact groups*, Invent. Math. **100** (1990), no. 2, 383–401, [doi:10.1007/BF01231192](#).
- [BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008.
- [BV] M. E. B. Bekka and A. Valette, *Kazhdan’s property (T) and amenable representations*, Math. Z. **212** (1993), no. 2, 293–299, [doi:10.1007/BF02571659](#).
- [BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^* -algebras*, Math. Ann. **307** (1997), no. 3, 343–380, [doi:10.1007/s002080050039](#).
- [CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^* -algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, [doi:10.1016/j.jfa.2013.04.004](#), [arXiv:1210.4050](#).
- [CLV] M. D. E. Conder, G. Liversidge, and M. R. Vsemirnov, *Generating pairs for $\mathrm{SL}(n, \mathbb{Z})$* , J. Algebra **662** (2025), 123–137, [doi:10.1016/j.jalgebra.2024.08.008](#).
- [CRW] M. D. E. Conder, E. F. Robertson, and P. Williams, *Presentations for 3-dimensional special linear groups over integer rings*, Proc. Amer. Math. Soc. **115** (1992), no. 1, 19–26, [doi:10.2307/2159559](#).
- [Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K -theory*, Adv. Math. **384** (2021), Paper No. 107722, [doi:10.1016/j.aim.2021.107722](#).
- [DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, [doi:10.1017/fms.2020.5](#).
- [ES05] G. Elek and E. Szabó, *Hyperlinearity, essentially free actions and L^2 -invariants: the sofic property*, Math. Ann. **332** (2005), no. 2, 421–441, [doi:10.1007/s00208-005-0640-8](#), [arXiv:math/0408400](#).

- [ES06] G. Elek and E. Szabó, *On sofic groups*, J. Group Theory **9** (2006), no. 2, 161–171, [doi:10.1515/JGT.2006.011](#), [arXiv:math/0305352](#).
- [FFF] F. Fournier-Facio, *A torsion-free non-sofic group*, preprint, 2026, [arXiv:2608.02025](#).
- [FGH] A. Fox, I. Goldbring, and B. Hart, *Locally universal C^* -algebras with computable presentations*, J. Funct. Anal. **287** (2024), no. 12, Paper No. 110652, [doi:10.1016/j.jfa.2024.110652](#), [arXiv:2303.02301](#).
- [Fri] T. Fritz, *On infinite-dimensional state spaces*, J. Math. Phys. **54** (2013), no. 5, 052107, [doi:10.1063/1.4807079](#).
- [GKEMP] D. Gao, S. Kunnawalkam Elayavalli, A. Manzoor, and G. Patchell, *A new source of purely finite matricial fields*, preprint, 2026, [arXiv:2603.24502](#).
- [Gl] L. Glebsky, *Extension of a residually finite group by a residually finite group is weakly sofic*, preprint, 2019, [arXiv:1910.08631](#).
- [GR] L. Glebsky and L. M. Rivera, *Sofic groups and profinite topology on free groups*, J. Algebra **320** (2008), no. 9, 3512–3518, [doi:10.1016/j.jalgebra.2008.08.008](#).
- [GHW] E. Guentner, N. Higson, and S. Weinberger, *The Novikov conjecture for linear groups*, Publ. Math. Inst. Hautes Études Sci. **101** (2005), 243–268, [doi:10.1007/s10240-005-0030-5](#).
- [HT] U. Haagerup and S. Thorbjørnsen, *A new application of random matrices: $\text{Ext}(C_{\text{red}}^*(F_2))$ is not a group*, Ann. of Math. (2) **162** (2005), no. 2, 711–775, [doi:10.4007/annals.2005.162.711](#), [arXiv:math/0212265](#).
- [JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $\text{MIP}^* = \text{RE}$, Commun. ACM **64** (2021), no. 11, 131–138, [doi:10.1145/3485628](#), [arXiv:2001.04383](#).
- [KW1] E. Kirchberg and S. Wassermann, *Exact groups and continuous bundles of C^* -algebras*, Math. Ann. **315** (1999), no. 2, 169–203, [doi:10.1007/s002080050364](#).
- [KW2] E. Kirchberg and S. Wassermann, *Permanence properties of C^* -exact groups*, Doc. Math. **4** (1999), 513–558, [doi:10.4171/DM/67](#).
- [Ku16] G. Kun, *On sofic approximations of property (T) groups*, preprint, 2016, [arXiv:1606.04471](#).
- [KT19] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan’s property (T)*, preprint, 2019, [arXiv:1901.03963](#).
- [KT26] G. Kun and A. Thom, *Nonsofic wreath products of residually finite groups*, preprint, 2026, [arXiv:2608.06222](#).
- [Lan] E. C. Lance, *On nuclear C^* -algebras*, J. Funct. Anal. **12** (1973), 157–176.
- [Lo] T. A. Loring, *Lifting solutions to perturbing problems in C^* -algebras*, Fields Institute Monographs, vol. 8, American Mathematical Society, Providence, RI, 1997.
- [Lo98] T. A. Loring, *C^* -algebras that are only weakly semiprojective*, Proc. Amer. Math. Soc. **126** (1998), no. 9, 2713–2715, [doi:10.1090/S0002-9939-98-04292-0](#).
- [LuO] A. Lubotzky and I. Oppenheim, *Non p -norm approximated groups*, preprint, 2018, [arXiv:1807.06790](#).
- [Mal] A. I. Mal’cev, *On isomorphic matrix representations of infinite groups*, Mat. Sb. **8(50)** (1940), 405–422; English translation, Amer. Math. Soc. Transl. (2) **45** (1965), 1–18.
- [MdIS] M. Magee and M. de la Salle, *$\text{SL}_4(\mathbb{Z})$ is not purely matricial field*, C. R. Math. Acad. Sci. Paris **362** (2024), 903–910, [doi:10.5802/crmath.617](#), [arXiv:2312.03220](#).
- [NST] N. Nikolov, J. Schneider, and A. Thom, *Some remarks on finitarily approximable groups*, J. Éc. polytech. Math. **5** (2018), 239–258, [doi:10.5802/jep.69](#).
- [Ne] B. H. Neumann, *Some remarks on infinite groups*, J. London Math. Soc. **12** (1937), 120–127, [doi:10.1112/jlms/s1-12.46.120](#).

- [OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, <https://cdn.openai.com/pdf/ten-proofs-oai.pdf>.
- [Ra] M. O. Rabin, *Recursive unsolvability of group theoretic problems*, Ann. of Math. (2) **67** (1958), 172–194.
- [Sc] C. Schafhauser, *Finite dimensional approximations of certain amalgamated free products of groups*, preprint, 2023, [arXiv:2306.02498](https://arxiv.org/abs/2306.02498).
- [Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, [arXiv:2603.13564](https://arxiv.org/abs/2603.13564).
- [Slo17] W. Slofstra, *The set of quantum correlations is not closed*, Forum Math. Pi **7** (2019), Paper No. e1, [doi:10.1017/fmp.2018.3](https://doi.org/10.1017/fmp.2018.3).
- [Slo18] W. Slofstra, *A group with at least subexponential hyperlinear profile*, preprint, 2018, [arXiv:1806.05267](https://arxiv.org/abs/1806.05267).
- [SV] W. Slofstra and T. Vidick, *Entanglement in non-local games and the hyperlinear profile of groups*, Ann. Henri Poincaré **19** (2018), no. 10, 2979–3005, [doi:10.1007/s00023-018-0718-y](https://doi.org/10.1007/s00023-018-0718-y).
- [Th] A. Thom, *Finitary approximations of groups and their applications*, Proceedings of the International Congress of Mathematicians — Rio de Janeiro 2018, [arXiv:1712.01052](https://arxiv.org/abs/1712.01052).
- [TWW] A. Tikuisis, S. White, and W. Winter, *Quasidiagonality of nuclear C^* -algebras*, Ann. of Math. (2) **185** (2017), no. 1, 229–284, [doi:10.4007/annals.2017.185.1.4](https://doi.org/10.4007/annals.2017.185.1.4), [arXiv:1509.08318](https://arxiv.org/abs/1509.08318).